

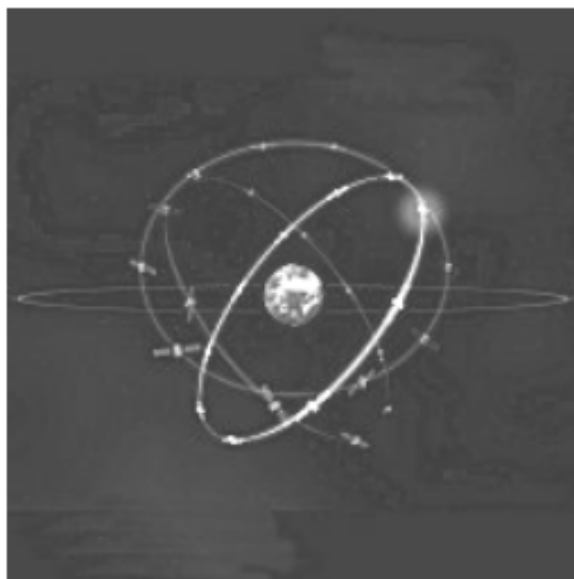


Image: ESA

Figure 1: .

MASALA 1. GEOSINXRON SUN'IY YO'LDOSH

IPhO 2005, 1-masala

Koinot apparatlari tomonidan eng ko'p bajariladigan orbital manevrlar parvoz yo'nalishi bo'ylab tezlik o'zgarishlaridan, ya'ni yuqori orbitalarga chiqish uchun tezlanishlar yoki atmosferaga qayta kirishni boshlash uchun tormozlanishlardan iborat. Ushbu masalada biz dvigatel kuchi radial yo'nalishda qo'llanilganda orbital o'zgarishlarni o'rganamiz. Raqamli qiymatlarni olish uchun quyidagilardan foydalaning: Yer radiusi $R_T = 6,37 \cdot 10^6$ m, Yer sirtidagi tortishish tezlanishi $g = 9,81$ m/s², va yulduz kunining uzunligi $T_0 = 24,0$ h.

Biz massasi m bo'lgan geosinxron¹ aloqa sun'iy yo'ldoshini r_0 radiusli ekvatorial aylanma orbitada joylashtirilgan deb qaraymiz. Ushbu sun'iy yo'ldoshlarda yakuniy orbitaga chiqish uchun zarur bo'lgan urinma kuchlarni ta'minlovchi "apogey dvigateli" mavjud.

Ballar har bir kichik savolning boshida, qavs ichida ko'rsatilgan.

1.1

(0.3 ball)

r_0 ning son qiymatini hisoblang.

1.2

(0.3+0.1 ball)

Sun'iy yo'ldoshning v_0 tezligining g , R_T va r_0 orqali analitik ifodasini keltiring va uning son qiymatini hisoblang.

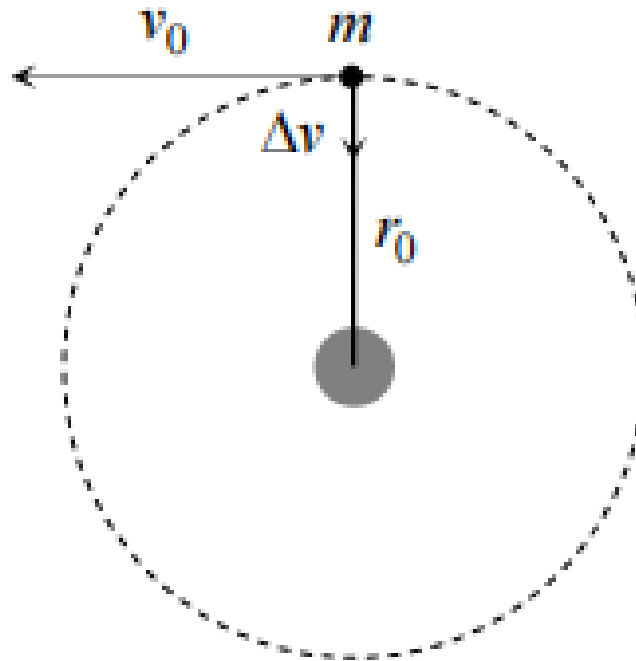
¹Geosinxron orbita — Yer bilan bir xil aylanish davriga ega bo'lgan orbita.

1.3

(0.4+0.4 ball)

Uning burchakli impulsi L_0 va mexanik energiyasi E_0 ning v_0 , m , g va R_T orqali ifodalarini oling.

Ushbu geosinxron aylanma orbitaga erishilgandan so'ng (F-1-rasmga qarang), sun'iy yo'ldosh kerakli joyda barqarorlashtirilgan va o'z ishini bajarishga tayyorlanayotganda, yer ustidan boshqaruvchilarning xatosi tufayli apogey dvigateli yana ishga tushiriladi. Kuch Yerga qarab yo'nalgan bo'ladi va yer ekipajining dvigatelni o'chirishga tez munosabatiga qaramay, sun'iy yo'ldoshga istalmagan tezlik o'zgarishi Δv beriladi. Biz bu turtki (boost) ni $\beta = \Delta v/v_0$ parametri bilan tavsiflaymiz. Dvigatelning yonish davomiyligi har qanday boshqa orbital vaqtlarga nisbatan har doim ahamiyatsiz, shuning uchun uni oniy deb hisoblash mumkin.



F-1

Figure 2: F-1-rasm: Geosinxron orbita va dvigatelning noto'g'ri ishga tushishi.

2-savol

$\beta < 1$ deb faraz qiling.

2.1

(0.4+0.5 ball)

Yangi orbitaning parametrlarini — yarim latus rektum l va eksentrisitet ε ni r_0 va β orqali aniqlang.

2.2**(1.0 ball)**

Yangi orbitaning katta o'qi bilan tasodifiy noto'g'ri ishga tushirish joyidagi radius-vektor orasidagi α burchagini hisoblang.

2.3**(1.0+0.2 ball)**

Perigey $r_{\min}$ va apogey $r_{\max}$ masofalarining Yer markazigacha bo'lgan analitik ifodalarini r_0 va β orqali keltiring va $\beta = 1/4$ uchun ularning son qiymatlarini hisoblang.

2.4**(0.5+0.2 ball)**

Yangi orbitaning davri T ni T_0 va β orqali ifodalang va $\beta = 1/4$ uchun uning son qiymatini hisoblang.

3-savol**3.1****(0.5 ball)**

Sun'iy yo'ldoshning Yer tortishishidan qochishi uchun zarur bo'lgan minimal turtki parametri β_{esc} ni hisoblang.

3.2**(1.0 ball)**

Bu holda yangi traektoriyada sun'iy yo'ldoshning Yer markaziga eng yaqin kelish masofasi $r'_{\min}$ ni r_0 orqali aniqlang.

4-savol

$\beta > \beta_{\text{esc}}$ deb faraz qiling.

4.1**(1.0 ball)**

Cheksizlikdagi qoldiq tezlikni v_{∞} ni v_0 va β orqali aniqlang.

4.2**(1.0 ball)**

Asimptotik qochish yo'nalishining "ta'sir parametri" (impact parameter) b ni r_0 va β orqali oling. (F-2-rasmga qarang).

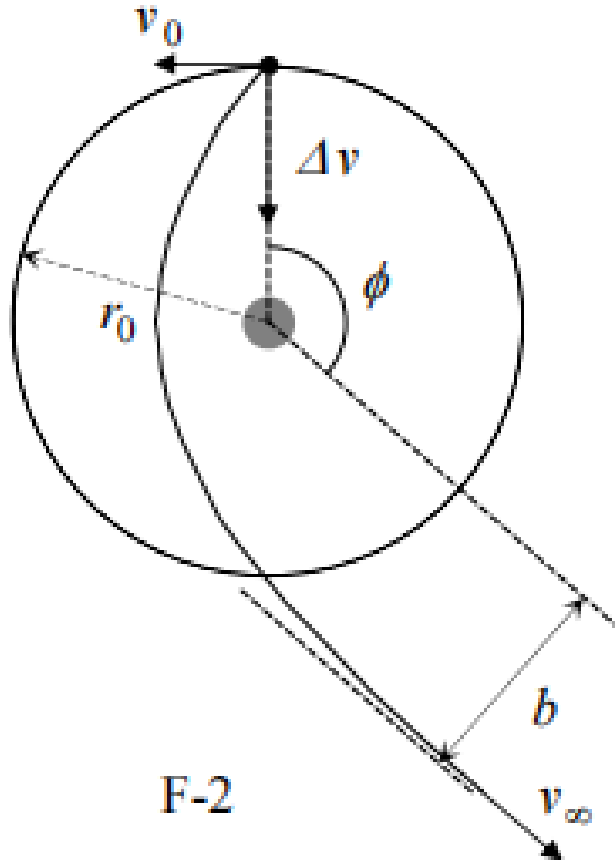


Figure 3: F-2-rasm: Asimptotik qochish trayektoriyasi.

4.3

(1.0+0.2 ball)

Asimptotik qochish yo'nalishining ϕ burchagini β orqali aniqlang. Uning son qiymatini $\beta = \frac{3}{2}\beta_{\text{esc}}$ uchun hisoblang.

MASLAHAT

Markaziy kuchlar ta'sirida, teskari kvadrat qonuniga bo'ysungan holda, jismlar ellips, parabola yoki giperbola bo'yicha harakatlanadi. $m \ll M$ yaqinlashishida tortuvchi massa M fokuslardan birida joylashgan. Shuning uchun bu fokusni koordinata boshi deb olib, bunday egri chiziqlarning umumiy qutb tenglamasi (F-3-rasmga qarang) quyidagicha yoziladi:

$$r(\theta) = \frac{l}{1 - \varepsilon \cos \theta}$$

bu yerda l — musbat doimiy (yarim latus rektum), ε — egri chiziqning eksentrisiteti. Harakat doimiylari orqali:

$$l = \frac{L^2}{GMm^2}, \quad \varepsilon = \left(1 + \frac{2EL^2}{G^2M^2m^3}\right)^{1/2}$$

bu yerda G — Nyuton doimiysi, L — aylanayotgan jismning bosh nuqtaga nisbatan burchakli impulsining moduli, E — uning mexanik energiyasi (cheksizlikda potensial energiya nol deb olingan).

Quyidagi holatlar mumkin:

1. Agar $0 \leq \varepsilon < 1$ bo'lsa, egri chiziqli ellips ($\varepsilon = 0$ da aylana).
2. Agar $\varepsilon = 1$ bo'lsa, egri chiziqli parabola.
3. Agar $\varepsilon > 1$ bo'lsa, egri chiziqli giperbola.

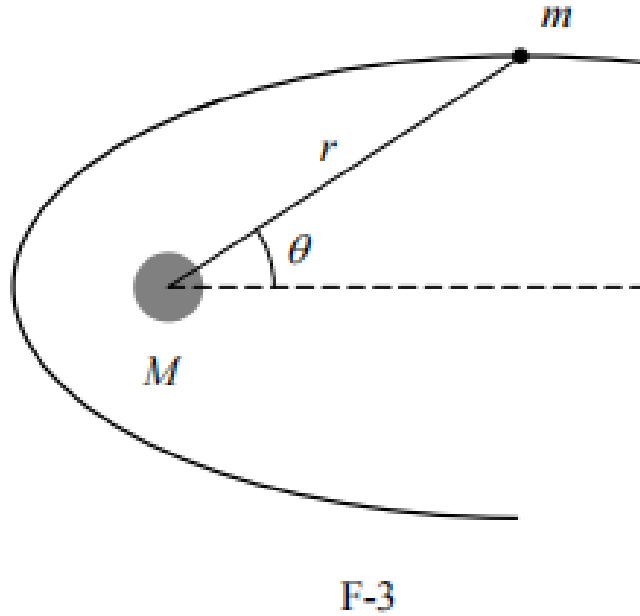


Figure 4: F-3-rasm: Qutb koordinatalarida konus kesimlari.

1-MASALA JAVOB VARAQASI

Savol	Asosiy formulalar va g'oyalar	Analitik natijalar	Raqamli natijalar	Ballar
1.1			$r_0 =$	0.3
1.2		$v_0 =$	$v_0 =$	0.4
1.3		$L_0 =$		0.4
		$E_0 =$		0.4
2.1		$l =$		0.4
		$\varepsilon =$		0.5
2.2			$\alpha =$	1.0
2.3		$r_{\max} =$	$r_{\max} =$	1.2
		$r_{\min} =$	$r_{\min} =$	
2.4		$T =$	$T =$	0.7
3.1			$\beta_{\text{esc}} =$	0.5
3.2		$r'_{\min} =$		1.0
4.1		$v_{\infty} =$		1.0
4.2		$b =$		1.0
4.3		$\phi =$	$\phi =$	1.2

2-MASALA. ELEKTR KATTALIKLARNING MUTLAQ O'LCHOVLARI

XIX asrda sodir bo'lgan texnologik va ilmiy o'zgarishlar elektr kattaliklari uchun umume'tirof etilgan standartlarga bo'lgan kuchli ehtiyojni keltirib chiqardi. Yangi mutlaq birliklar faqat Fransuz inqilobidan keyin o'rnatilgan uzunlik, massa va vaqt standartlariga tayanishi kerak deb o'ylangan. Ushbu birliklarning qiymatlarini aniqlash uchun 1861 yildan 1912 yilgacha intensiv eksperimental ish olib borildi. Biz bu erda uchta amaliy tadqiqotni taklif qilamiz.

Ballar har bir kichik savolning boshida, qavs ichida ko'rsatilgan.

Om qarshiligini aniqlash (Kelvin)

N o'ramli, radiusi a va umumiy qarshiligi R bo'lgan yopiq aylana g'altak gorizontali magnit maydoni $\vec{B}_0 = B_0 \vec{i}$ ichida vertikal diametri atrofida bir tekis burchak tezligi ω bilan aylantirilmoqda.

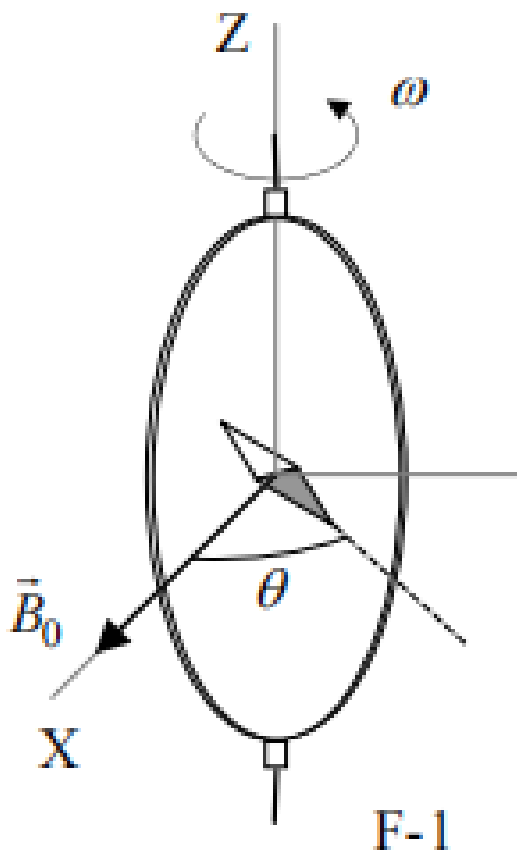


Figure 5: F-1-rasm: Aylanuvchi g'altak va magnit ignasi.

1. $(0.5+1.0 \text{ ball})$ G'altakda induktsiyalangan elektrik yurituvchi kuch $\mathcal{E}$

ni va g'altakni harakatda ushlab turish uchun zarur bo'lgan o'rtacha quvvat $\langle P \rangle$ ni hisoblang. G'altakning o'zinduksiyasini e'tiborsiz qoldiring.

Kichik magnit ignasi g'altakning markaziga joylashtirilgan (F-1-rasmda ko'rsatilgan). U Z o'qi atrofida gorizontaal tekislikda sekin aylanishi mumkin, lekin g'altakning tez aylanishiga ergasha olmaydi.

- 2 (2.0 ball) Turg'un rejimga erishilgandan so'ng, igna $\vec{B}_0$ bilan kichik θ burchak hosil qiladigan yo'nalishda o'rnashadi. G'altakning qarshiligi R ni ushbu burchak va tizimning boshqa parametrlari orqali ifodalang.

Lord Kelvin 1860-yillarda omning mutlaq standartini o'rnatish uchun ushbu usuldan foydalangan. Aylanuvchi g'altakdan qochish uchun Lorenz muqobil usulni ishlab chiqdi, undan Lord Rayleigh va Ms. Sidgwick foydalangan. Biz quyidagi paragraflarda ushbu usulni tahlil qilamiz.

Om qarshiligini aniqlash (Rayleigh, Sidgwick)

Tajriba qurilmasi F-2-rasmda ko'rsatilgan. U SS o'tkazuvchan valga o'rnatilgan ikkita bir xil metall disk D va D' dan (radiusi b) iborat. Motor to'plamni R qarshilikni o'lchash uchun sozlanishi mumkin bo'lgan ω burchak tezligi bilan aylantiradi. Ikkita bir xil g'altak C va C' (radiusi a , har biri N o'ramli) disklarni o'rab turadi. Ular shunday ulanganki, tok I ular orqali qarama-qarshi yo'nalishlarda oqadi. Butun qurilma R qarshilikni o'lchashga xizmat qiladi.

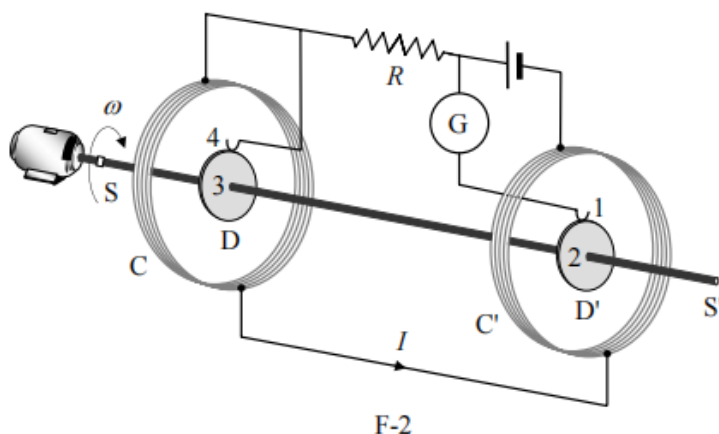


Figure 6: F-2-rasm: Rayleigh–Sidgwick qurilmasi.

- 3 (2.0 ball) C va C' g'altaklaridan oqayotgan I toki D va D' atrofida bir jinsli magnit maydon B ni hosil qiladi deb faraz qiling (bu maydon g'altak markazidagi maydonga teng). Disklarning aylanish o'qi ularning markazlaridan o'tadi. Chetki nuqtalar 1 va 4 oralig'ida induktsiyalangan elektrik yurituvchi kuch ε ni hisoblang. G'altaklar orasidagi masofa g'altak radiusidan ancha katta va $a \gg b$ deb qabul qiling.

Disklar zanjirga ularning 1 va 4 chetki nuqtalarida cho'tka kontaktlar orqali ulangan. Galvanometr G 1-2-3-4 zanjiri bo'ylab tok oqimini qayd etadi.

- 4 (0.5 ball) G nol ko'rsatganda R qarshilik o'lchanadi. R ni tizimning fizik parametrlari orqali ifodalang.

Ampelni aniqlash

Ikki o'tkazgichdan tok o'tkazish va ular orasidagi kuchni o'lchash tokning mutlaq qiymatini aniqlash imkonini beradi. Lord Kelvin tomonidan 1882 yilda ishlab chiqilgan "Tok balans" ushbu usuldan foydalanadi. U ketma-ket ulangan a radiusli oltita bir xil bir o'ramli g'altaklardan $C_1 \dots C_6$ iborat. F-3-rasmda ko'rsatilganidek, qo'zg'almas g'altaklar C_1, C_3, C_4 va C_6 kichik $2h$ masofa bilan ajratilgan ikkita gorizontall tekislikda joylashgan. C_2 va C_5 g'altaklari d uzunlikdagi muvozanat qo'llariga o'rnatilgan va muvozanat holatida ikkala tekislikdan teng masofada joylashgan.

Tok I turli g'altaklardan shunday yo'nalishda oqadiki, C_2 ga magnit kuchi yuqoriga, C_5 ga esa pastga yo'nalgan bo'ladi. Tok zanjir orqali oqganda yuqorida tavsiflangan muvozanat holatini tiklash uchun tayanch nuqtasi O dan x masofada joylashgan m massali og'irlik talab qilinadi.

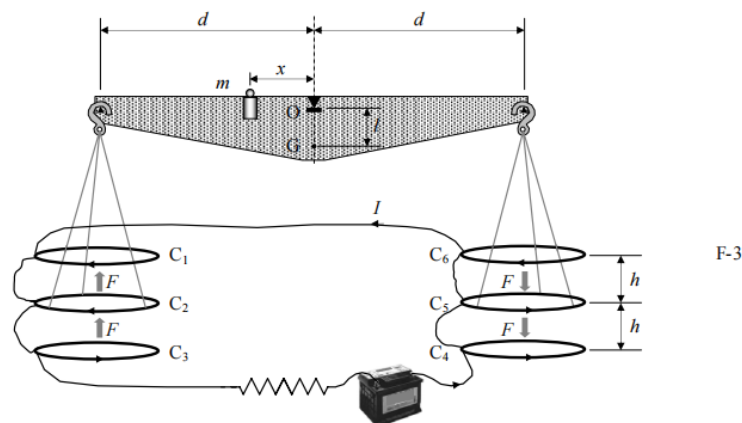


Figure 7: F-3-rasm: Tok balans sxemasi.

- 5 (1.0 ball) C_2 ga C_1 bilan magnit o'zaro ta'siri tufayli ta'sir qiluvchi F kuchni hisoblang. Soddashtirish uchun, birlik uzunlikka to'g'ri keladigan kuch parallel toklar o'tayotgan ikkita uzun to'g'ri simlarga mos keladi deb faraz qiling.
- 6 (1.0 ball) Balans muvozanat holatida bo'lganda I toki o'lchanadi. I ni tizimning fizik parametrlari orqali ifodalang. Qurilmaning o'lchamlari shundayki, chap va o'ng tomondagi g'altaklarning o'zaro ta'sirini e'tiborsiz qoldirish mumkin.

M balansning massasi (m va osma qismlardan tashqari), G uning massa markazi va l masofa $\overline{OG}$ bo'lsin.

- 7 (2.0 ball) Balans muvozanati C_2 balandligining kichik δz va C_5 balandligining $-\delta z$ o'zgarishiga olib keladigan og'ishlarga nisbatan barqarordir. Balans qo'yib yuborilganda ham muvozanat holatiga qaytishi uchun $\delta z_{\max}$ maksimal qiymatini hisoblang.

Eslatmalar:

- G'altaklarning markazlari taxminan bir chiziqda qoladi deb hisoblang.

- $\beta \ll 1$ uchun $\frac{1}{1 \pm \beta} \approx 1 \mp \beta + \beta^2$ yoki $\frac{1}{1 \pm \beta^2} \approx 1 \mp \beta^2$ va kichik θ uchun $\sin \theta \approx \tan \theta$ yaqinlashishlaridan foydalaning.

2-MASALA JAVOB VARAQASI

Savol	Qo'llanilgan asosiy formulalar	Analitik natijalar	Ballar
1		$\mathcal{E} =$	1.5
		$\langle P \rangle =$	
2		$R =$	2.0
3		$\varepsilon =$	2.0
4		$R =$	0,5
5		$F =$	1.0
6		$I =$	1.0
7		$\delta z_{\max} =$	2.0

3-MASALA. GRAVITATSIYON MAYDONDAGI NEYTRONLAR

IPhO 2005, 3-masala

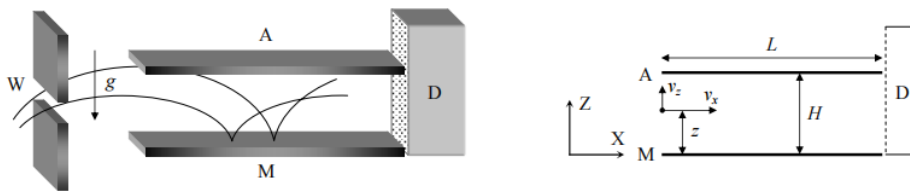
Tanish klassik dunyoda, Yer yuzasida elastik sakrayotgan to'p abadiy harakatning ideal namunasidir. To'p tuzoqqa tushgan: u sirtidan pastga yoki o'zining burilish nuqtasidan yuqoriga chiqolmaydi. U shu holatda chegaralangan holda qoladi, qayta-qayta pastga tushib va yuqoriga sakrab, abadiy. Faqat havo qarshiligi yoki noelastik sakrashlar jarayonni to'xtatishi mumkin va quyida ular e'tiborsiz qoldiriladi.

2002 yilda Grenobldagi Laue-Langevin institutidan bir guruh fiziklar neytronlarning Yer tortishish maydonidagi harakati haqida eksperimental dalillar haqida xabar berishdi. Tajribada o'ngga harakatlanayotgan neytronlar neytron oynasi vazifasini o'taydigan gorizontall kristall sirt tomon tushishga ruxsat etilgan va u erda ular yana boshlang'ich balandlikka qadar elastik ravishda qaytgan.

Tajriba qurilmasi F-1-rasmda tasvirlangan. U W teshigi, M neytron oynasi ($z = 0$ balandlikda), A neytron yutgichi ($z = H$ balandlikda va L uzunlikda) va D neytron detektoridan iborat. Neytronlar dastasi W dan D gacha A va M orasidagi bo'shliq orqali o'zgarimas gorizontall tezlik komponenti v_x bilan uchadi. A sirtiga yetib borgan barcha neytronlar yutiladi va tajribadan yo'qoladi. M sirtiga yetib borganlar esa elastik qaytariladi. D detektori o'tkazish tezligi $N(H)$ ni, ya'ni D ga vaqt birligida yetib keladigan neytronlarning umumiy sonini hisoblaydi.

Ballar har bir kichik savolning boshida, qavs ichida ko'rsatilgan.

Neytronlar bo'shliqqa vertikal tezliklarning v_z keng diapazonida (musbat va manfiy) kiradi. Bo'shliqqa kirgandan so'ng ular pastdagi oyna va yuqoridagi yutgich orasida uchadi.



F-1

Figure 8: F-1-rasm: Tajriba qurilmasi sxemasi.

1

(1.5 ball)

Klassik ravishda, z balandlikda kirgan neytronlarning detektor D ga yetib borishi mumkin bo'lgan vertikal tezliklari $v_z(z)$ oralig'ini hisoblang. L masala boshqa barcha uzunliklardan ancha katta deb faraz qiling.

2

(1.5 ball)

Oldingi tezlik oralig'idan tashqaridagi barcha neytronlar, z qiymatlaridan qat'i nazar, A tomonidan yutilishini ta'minlash uchun bo'shliqning minimal uzunligi L_c ni klassik ravishda hisoblang. $v_x = 10$ m/s va $H = 50$ μ m dan foydalaning.

3

(2.5 ball)

Neytronlar bo'shliqqa v_z vertikal tezlik va z balandlik bilan kirishi va v_z va z ning barcha qiymatlari bir xil ehtimolli deb faraz qilib, klassik o'tkazish tezligi $N_c(H)$ ni hisoblang. Javobni ρ — vaqt birligida, vertikal tezlik birligida va balandlik birligida bo'shliqqa v_z vertikal tezlik va z balandlik bilan kiradigan neytronlarning doimiy soni orqali ifodalang.

Grenobl guruhi tomonidan olingan eksperimental natijalar yuqoridagi klassik bashoratlarga mos kelmaydi, aksincha $N(H)$ qiymati H ba'zi kritik balandliklar $H_1, H_2, \dots$ dan o'tganda keskin o'sishlarni boshdan kechirishini ko'rsatdi (F-2-rasmda eskiz). Boshqacha aytganda, tajriba oynada sakrayotgan neytronlarning vertikal harakati kvantlanganligini ko'rsatdi. Bor va Sommerfeld vodorod atomining energiya sathlarini olishda ishlatgan tilda buni quyidagicha yozish mumkin: "Neytronlarning vertikal yo'nalish bo'ylab ta'siri (action) S Plank ta'sir doimiysi h ning butun karralisidir". Bu erda S quyidagicha beriladi:

$$S = \int p_z(z) dz = nh, \quad n = 1, 2, 3, \dots \quad (\text{Bor-Sommerfeld kvantlash qoidasi})$$

bu yerda p_z — klassik impulsning vertikal komponenti va integral butun bir sakrash siklini qamraydi. Faqat S ning shunday qiymatlariga ega neytronlarga bo'shliqda ruxsat etiladi.

4

(2.5 ball)

Bor-Sommerfeld kvantlash shartidan foydalanib, burilish balandliklari H_n va (vertikal harakatga bog'liq) energiya sathlari E_n ni hisoblang. H_1 uchun μ m da va E_1 uchun eV da son natijalarni keltiring.

Neytronlarning kirishdagi bir tekis boshlang'ich taqsimoti ρ uzun bo'shliq orqali uchish davomida D da qayd etilgan pog'onali taqsimotga o'zgaradi (F-2-rasmga qarang). Bundan buyon, soddalashtirish uchun $H < H_2$ bo'lgan uzun bo'shliq holatini ko'rib chiqamiz. Klassik ravishda 1-savolda ko'rib chiqilgan diapazondagi barcha neytronlarga ruxsat etilgan bo'lsa, kvant mexanikasida faqat E_1 energiya sathidagi neytronlarga ruxsat beriladi. Heisenberg vaqt-energiya noaniqlik printsipiga ko'ra, bu qayta taqsimlanish minimal uchish vaqtini talab qiladi. Agar bo'shliq uzunligi kichik bo'lsa, vertikal harakat energiyasining noaniqligi sezilarli bo'ladi. Bu hodisa energiya sathlarining kengayishiga olib keladi.

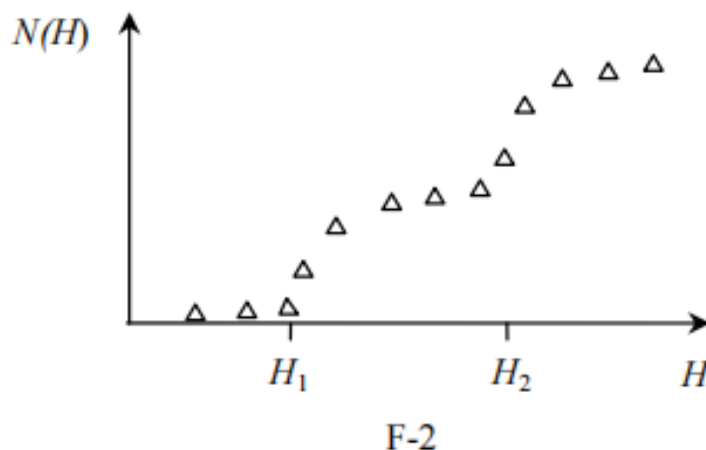


Figure 9: F-2-rasm: Neytron o'tkazish tezligining H ga bog'liqligi (eskiz).

5

(2.0 ball)

D da neytronlar sonining birinchi keskin o'sishini kuzatish uchun zarur bo'lgan minimal uchish vaqti t_q va bo'shliqning minimal uzunligi L_q ni baholang. $v_x = 10 \text{ m/s}$ dan foydalaning.

Ma'lumotlar

Parametr	Qiymat
Plank ta'sir doimiysi	$h = 6,63 \cdot 10^{-34} \text{ J}\cdot\text{s}$
Vakuumdagi yorug'lik tezligi	$c = 3,00 \cdot 10^8 \text{ m/s}$
Elementar zaryad	$e = 1,60 \cdot 10^{-19} \text{ C}$
Neytron massasi	$M = 1,67 \cdot 10^{-27} \text{ kg}$
Yerdagi tortishish tezlanishi	$g = 9,81 \text{ m/s}^2$
Agar kerak bo'lsa, quyidagi ifodadan foydalaning:	$\int (1-x)^{1/2} dx = -\frac{2}{3}(1-x)^{3/2}$

3-MASALA JAVOB VARAQASI

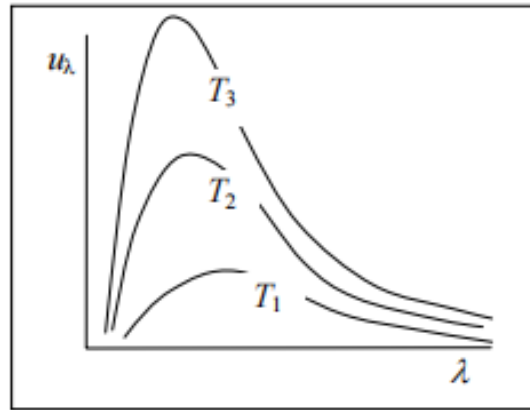
Savol	Qo'llanilgan asosiy formulalar	Analitik natijalar	Raqamli natijalar	Ballar
1	$\frac{1}{2}Mv_z^2 + Mgz \leq MgH$			1,5
2	$L_c = v_x \cdot 2t_f, H = \frac{1}{2}gt_f^2$			1,5
3	$\frac{dN_c}{dz} = \rho(v_{z,\max} - v_{z,\min})$			2,5
4	$S = 2 \int_0^H p_z dz = nh$			2,5
5	$\Delta E \Delta t \geq \hbar, \Delta E \approx E_1, \Delta x = v_x \Delta t$			2,0

PLANK DOIMIYSINING AKKORD LAMPASI NURI YORDAMIDA ANIQLANISHI

1900 yilda Plank materiya tomonidan yorug'lik energiyasi $h\nu$ kvantlari shaklida chiqariladi degan gipotezani kiritdi. 1905 yilda Eynshteyn bu g'oyani kengaytirib, chiqarilgandan so'ng energiya kvanti yorug'lik kvanti (keyinchalik foton nomini olgan) sifatida butun holda qoladi deb taklif qildi. Oddiy yorug'lik har bir to'lqin frontida juda ko'p sonli fotonlardan tashkil topgan. Ular to'lqin ichida, xuddi donador moddada alohida atomlar kabi, niqoblangan holda qoladi, ammo h – Plank doimiysi – ularning mavjudligini ochib beradi. Ushbu tajribaning maqsadi Plank doimiysini o'lchashdir.

Jism nafaqat chiqaradi, balki tashqaridan kelayotgan nurlanishni yutishi ham mumkin. Qora jism deb, har qanday to'lqin uzunligi uchun unga tushayotgan barcha nurlanishni yuta oladigan jismga aytiladi. U to'liq nurlantiruvchidir. Elektromagnit nurlanishga nisbatan, qora jismlar hamma narsani yutadi, hech narsani qaytarmaydi va hamma narsani chiqaradi. Haqiqiy jismlar to'liq qora emas; jism tomonidan chiqarilgan energiyaning bir xil haroratdagi qora jism tomonidan chiqarilgan energiyaga nisbati emissivlik ε deb ataladi va odatda to'lqin uzunligiga bog'liq.

Plank T absolyut haroratdagi jism tomonidan λ to'lqin uzunlikdagi elektromagnit nurlanish shaklida chiqarilgan quvvat zichligini quyidagicha yozish mumkinligini topdi:



F-1

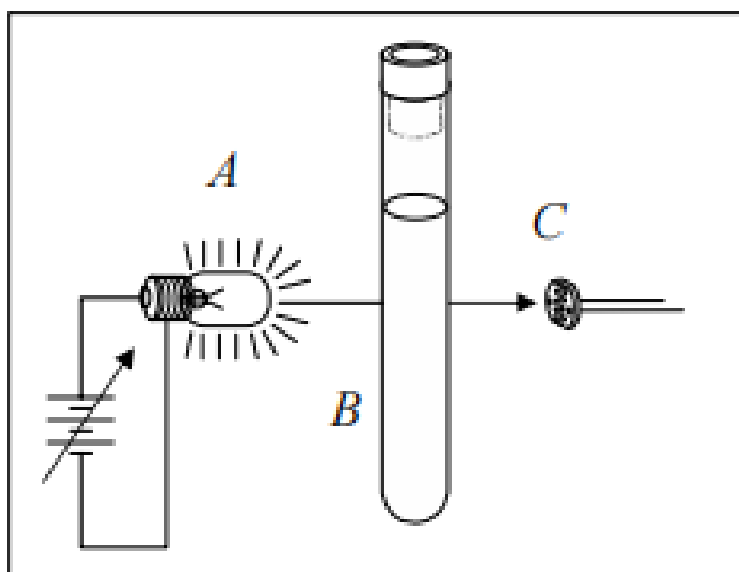
$$u_{\lambda} = \varepsilon \frac{c_1}{\lambda^5 (e^{c_2/(\lambda T)} - 1)} \quad (1)$$

bu yerda c_1 va c_2 konstantalardir. Ushbu savolda sizdan c_2 ni eksperimental ravishda aniqlashingiz so'raladi, u h ga proporsionaldir.

F-1-rasmdagi maksimumlarning chap tomonida, kichik λ larda chiqarish uchun (1) tenglamaning maxrajidagi -1 ni tashlab yuborish joizdir, bu esa quyidagiga qisqaradi:

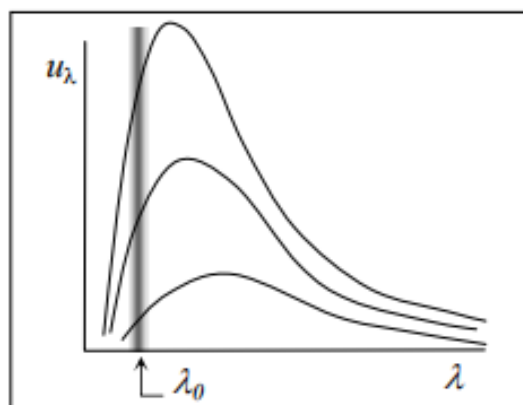
$$u_{\lambda} = \varepsilon \frac{c_1}{\lambda^5} e^{c_2/(\lambda T)} \quad (2)$$

Ushbu tajriba savolining asosiy elementlari F-2-rasmda chizilgan.



F-2

Figure 10: F-2-rasm: Tajriba qurilmasi sxemasi.



F-3

- Nurlantiruvchi jism — bu akkord lampasining volfram filamenti A bo'lib, u λ larning keng diapazonini chiqaradi va uning yorqinligi o'zgartirilishi mumkin.
- Probirka B faqat ko'rinadigan spektrning λ_0 qiymati atrofida tor bandini o'tkazuvchi suyuq filtrni o'z ichiga oladi (F-3-rasmga qarang). Filtr xossalariga oid qo'shimcha ma'lumot 5-betda keltirilgan.
- Nihoyat, o'tgan nurlanish fotoqarshilik C (LDR – Light Dependent Resistor qisqartmasi) ustiga tushadi. LDR ning ba'zi xossalari 6-betda tasvirlangan.

LDR qarshiligi R uning yoritilganligi E ga bog'liq bo'lib, u filamentning quvvat energiya zichligiga proporsionaldir:

$$E \propto u_{\lambda_0}, \quad R \propto E^{-\gamma}$$

bu erda γ o'lchamsiz parametr LDR ning xossasi bo'lib, tajribada aniqlanadi. Ushbu qurilma uchun biz LDR qarshiligi R va filament harorati T o'rtasidagi munosabatni olamiz:

$$R = c_3 e^{c_2/(\lambda_0 T)} \quad (3)$$

Biz bundan 6-betda foydalanamiz. Bu munosabatda c_3 noma'lum proporsionallik konstantasidir. R ni T ning funksiyasi sifatida o'lchab, c_2 ni olish mumkin, bu tajriba savolining maqsadidir.

QURILMANING TAVSIFI

Qurilmaning tarkibiy qismlari F-4-rasmda ko'rsatilgan, unda uni sozlash uchun ba'zi ko'rsatmalar ham berilgan. Barcha qismlar mavjudligini tekshiring, ammo keyingi sahifadagi ko'rsatmalarni o'qimaguningizcha ular bilan hech qanday manipulyatsiya qilishdan saqlaning.

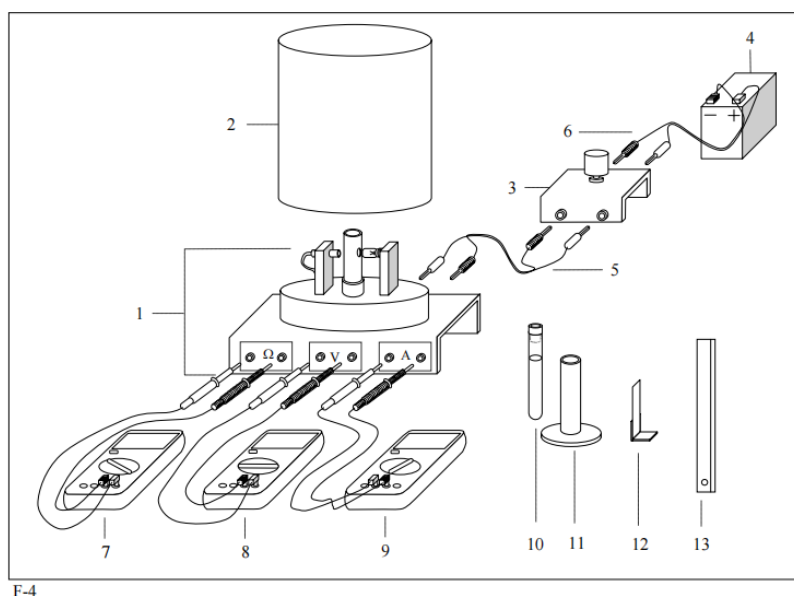


Figure 11: F-4-rasm: Qurilma sxemasi.

JIHAZLAR

1. Platforma. Uning tepasida LDR uchun tutqich, probirka uchun tutqich va 12 V, 0,1 A elektr lampasi uchun tutqichni ushlab turadigan disk mavjud.
2. Himoya qopqog'i.
3. 10 aylanishli va 1 k potensiometr.
4. 12 V batareya.

5. Platformani potentsiometrغا ulash uchun ikkala uchida vilkali qizil va qora simlar.
6. Bir uchida vilka va ikkinchi uchida batareya uchun rozetkalari bo'lgan qizil va qora simlar.
7. Ommetr sifatida ishlaydigan multimetr.
8. Voltmetr sifatida ishlaydigan multimetr.
9. Ampermetr sifatida ishlaydigan multimetr.
10. Suyuq filtrli probirka.
11. Probirka uchun stend.
12. Kulrang filtr.
13. Chizg'ich.

Multimetrlardan foydalanish bo'yicha qisqartirilgan ko'rsatmalar to'plami, shuningdek eng kichik kvadratlar usuli haqidagi ma'lumotlar alohida sahifada berilgan.

QURILMANI SOZLASH

Quyidagi ko'rsatmalarga amal qiling:

- F-4-rasmda ko'rsatilganidek, elektr ulanishlarni ehtiyotkorlik bilan bajaring, lekin simlarni 6-potentsiometrغا ulamang.
 - F-5-rasmga qarab, quyida ko'rsatilgan qadamlarni bajaring:
1. Potentsiometr tugmachasini soat miliga teskari yo'nalishda oxirigacha aylantiring.
 2. Probirka tutqichini asta-sekin aylantiring, shunda yon teshiklardan biri lampa qarshisida, ikkinchisi esa LDR qarshisida bo'lsin.
 3. LDR ni probirka tutqichiga uning yon teshigiga engil tegib turmaguncha yaqinlashtiring. LDR sirtini F-5-rasmda ko'rsatilgandek yo'naltirish tavsiya etiladi.
 4. Probirkani uning tutqichiga joylashtiring.
 5. Tashqi yorug'likdan himoya qilish uchun qopqoqni platformaga qo'ying. LDR qarshiligini o'lchashni boshlashdan oldin uni kamida 10 daqiqa davomida to'liq qorong'ulikda saqlashga ishonch hosil qiling. Bu ehtiyot chorasi, chunki qorong'ulikdagi qarshilik qiymatiga bir zumda erishilmaydi.

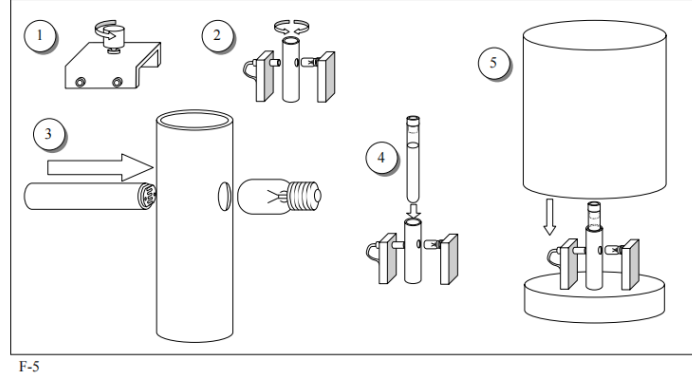


Figure 12: F-5-rasm: LDR va probirkaning joylashuvi.

1-topshiriq

1-javob varaqasida zanjir to'liq ulanganda, katakchalar ichida va katakchalar orasidagi to'liq elektr zanjirlarini chizing. Chizmalarni bajarishda F-4-rasmdagi ko'rsatmalarni hisobga oling.

Filament (qizdirish tolasining) haroratini o'lchash

O'tkazgich filamentning elektr qarshiligi R_B quyidagicha ifodalanadi:

$$R_B = \rho \frac{l}{S} \quad (4)$$

bu yerda ρ — o'tkazgichning solishtirma qarshiligi, l — uzunlik, S — filamentning ko'ndalang kesim yuzi.

Bu qarshilik haroratga bog'liq, turli sabablar tufayli, masalan:

- Metallning solishtirma qarshiligi harorat oshishi bilan ortadi. Volfram uchun va 300 K dan 3655 K gacha bo'lgan harorat oralig'ida XB birliklarida quyidagi empirik ifoda o'rinli:

$$T = 3,05 \cdot 10^8 \rho^{0,83} \quad (5)$$

- Termik kengayish filamentning uzunligi va kesimini o'zgartiradi. Biroq uning filament qarshiligiga ta'siri ushbu tajribada juda kichik bo'lib, uni e'tiborsiz qoldirish mumkin.

(4) va (5) ifodalardan va kengayishni e'tiborsiz qoldirib, quyidagi kelib chiqadi:

$$T = a R_B^{0,83} \quad (6)$$

Shuning uchun T ni topish uchun a ni aniqlash kerak. Bunga atrof-muhit harorati T_0 da filament qarshiligi $R_{B,0}$ ni o'lchash orqali erishiladi.

2-topshiriq

- a) Atrof-muhit harorati T_0 ni multimetr yordamida o'lchang.
- b) T_0 da filament qarshiligini $R_{B,0}$ o'lchash uchun ommetrdan foydalanish yaxshi fikr emas, chunki u filament haroratini oshiradigan kichik noma'lum tokni kiritadi. Buning o'rniga, $R_{B,0}$ ni topish uchun batareyani potensiometr ga ulang va eng past erishish mumkin bo'lgan kuchlanish qiymatlaridan boshlab 1 V gacha bo'lgan kuchlanishlar uchun yetarlicha ko'p tok ko'rsatkichlarini oling (100 mV dan pastda kamida 15 ko'rsatkich olish foydali bo'ladi). Oxirida potensiometrni boshlang'ich holatiga qaytaring va batareyadan potensiometr ga boradigan kabellardan birini uzing.

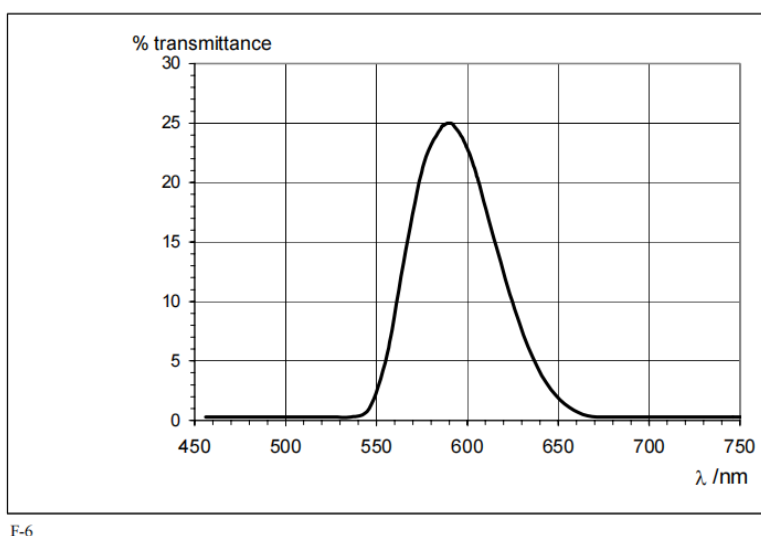
Har bir V va I juftligi uchun R_B ni toping va bu qiymatlarni 2-topshiriq, b) bandining javob varaqasidagi jadvalga ko'chiring. Eksperimental ravishda erisha olgan eng past kuchlanishni u erda ko'rsating. Grafik chizing va vertikal o'qda R_B ni, gorizontal o'qda I ni ifodalang.

- c) b) bandda olingan grafiklarni ko'rib chiqib, bosh nuqtasidagi ordinataga $R_{B,0}$ ekstrapolyatsiya qilish uchun mos chiziqli approksimatsiya qilish maqsadida qiymatlarning tegishli oralig'ini tanlang. Tanlangan qiymatlarni 2-topshiriq, c) bandining javob varaqasidagi jadvalga yozing. Nihoyat, $R_{B,0}$ va $\Delta R_{B,0}$ ni oling.
- d) (6) ifodadan foydalanib, $R_{B,0}$ (Ω da) va T_0 (K da) uchun a va Δa ning son qiymatlarini hisoblang.

FILTRNING OPTIK XOSSALARI

Probirkadagi suyuq filtr mis (II) sulfat va Orange (II) anilin bo'yog'ining suvli eritmasidir. Tuzning maqsadi filament chiqaradigan infraqizil nurlanishni yutishdir.

Filtrning o'tkazuvchanligi (o'tgan intensivlik / tushayotgan intensivlik) to'lqin uzunligiga bog'liq holda F-6-rasmda ko'rsatilgan.



F-6

Figure 13: F-6-rasm: Filtrning o'tkazuvchanligi.

3-topshiriq

F-6-rasmdan λ_0 va $\Delta\lambda$ ni aniqlang.

Eslatma: $2\Delta\lambda$ — balandlikning yarmidagi to'liq kenglik, λ_0 — maksimumdagi to'lqin uzunligi.

FOTOQARSHILIK (LDR) XOSSALARI

Fotoga qarshilik (LDR) ni tashkil etuvchi material qorong'ulik sharoitida o'tkazmaydi. Uni yoritish natijasida ba'zi zaryad tashuvchilar aktivlanadi va u orqali elektr toki oqishi mumkin. LDR qarshiligi uchun quyidagi munosabat yoziladi:

$$R = bE^{-\gamma} \quad (7)$$

bu yerda b — LDR ning tarkibi va geometriyasiga bog'liq doimiy, γ esa o'lchamsiz parametr bo'lib, qarshilikning tushayotgan nurlanishning E yoritilganligi bilan o'zgarishini o'lchaydi. Nazariy jihatdan ideal LDR uchun $\gamma = 1$, biroq ko'plab omillar ta'sir qiladi, shuning uchun haqiqiy holatda $\gamma < 1$.

γ ni aniqlash kerak. Bunga R va E juftligini o'lchash (F-7-rasm) va keyin lampa va probirka o'rtasiga o'tkazuvchanligi 51,2% deb ma'lum bo'lgan va xatosiz deb hisoblanadigan kulrang filtr F ni (F-8-rasm) kiritish orqali erishiladi. Bu $E' = 0,512E$ yoritilganlikni hosil qiladi. Ushbu yoritilganlikka mos keladigan R' qarshilikni o'lchab, quyidagiga ega bo'lamiz:

$$R = bE^{-\gamma}, \quad R' = b(0,512E)^{-\gamma}$$

Bundan

$$\ln \frac{R}{R'} = \gamma \ln 0,512 \quad (8)$$

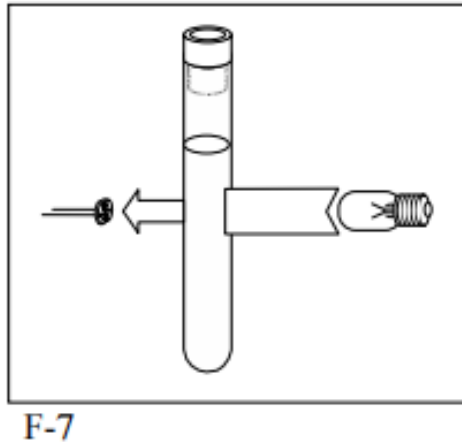


Figure 14: F-7-rasm: LDR ning yoritilganlikka bog'liqligi (sxematik).

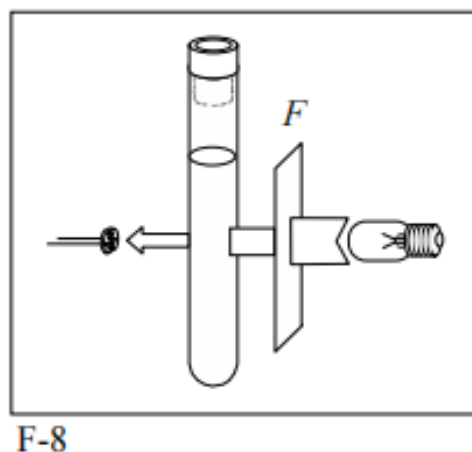
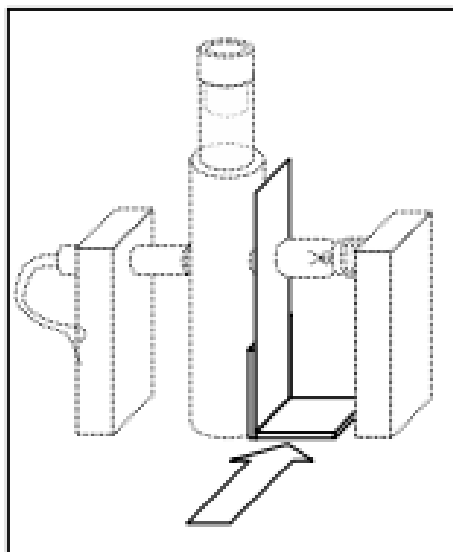


Figure 15: F-8-rasm: Kulrang filtr.

Ushbu amaliyotni quyidagi 4-topshiriqning b) bandiga kelguniga qadar bajarmang.

4-topshiriq

- a) Ushbu qismni boshlashdan oldin LDR kamida 10 daqiqa davomida to'liq qorong'ulikda qolganligini tekshiring. Batareyani potensimetrga ulang va tugmachani juda sekin aylantirib, lampaning kuchlanishini oshiring. 9,50 V dan 11,50 V gacha bo'lgan V oralig'ida V va I juftliklarini o'qing va mos LDR qarshiliklari R ni oling (kamida 12 ko'rsatkich olish foydali bo'ladi). Barcha qiymatlarni javob varaqasidagi jadvalga ko'chiring. LDR javobidagi kechikishni bartaraf etish uchun quyidagi amaliyotni tavsiya qilamiz: $V > 9,5$ V ga erishgandan so'ng, birinchi ko'rsatkichni olishdan oldin taxminan 10 daqiqa kuting. Keyin ikkinchi ko'rsatkich uchun 5 daqiqa kuting va hokazo. Har qanday hisob-kitobni bajarishdan oldin keyingi bosqichga o'ting.
- b) R qarshilikning eng kichik qiymatini olgach, himoya qopqog'ini oching, kulrang filtrni F-9-rasmda ko'rsatilgandek joylashtiring, platformani iloji boricha tezroq qaytadan yoping va yangi LDR qarshiligi R' ni qayd eting. Ushbu ma'lumotlardan (8) ifodada foydalanib, γ va $\Delta\gamma$ ni hisoblang.
- c) (3) tenglamani $\ln R$ ning $R_B^{-0,83}$ ga chiziqli bog'liqligini ko'rsatadigan qilib o'zgartiring. Ushbu tenglamani yozing va uni (9) deb belgilang.
- d) Endi a) banddagi ma'lumotlardan foydalanib, (9) tenglamani chizish uchun xizmat qiladigan jadval tuzing.
- e) Grafikni tuzing va $c_2 = hc/k$ ekanligini bilib, istalgan usul bilan h va Δh ni hisoblang (tashkilot tomonidan taqdim etilgan kalkulyatorlarning statistik funktsiyalaridan foydalanishga ruxsat beriladi).



F-9

Figure 16: F-9-rasm: Filtrni o'rnatish.

Ma'lumotlar:

Yorug'lik tezligi $c = 2,998 \cdot 10^8$ m/s;

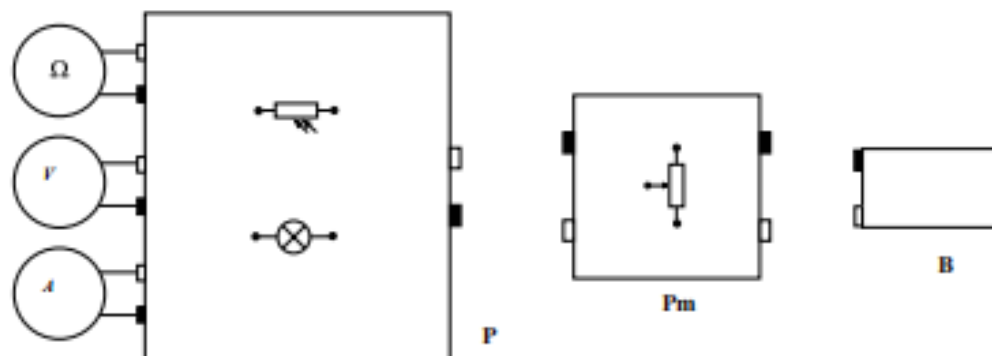
Bolsman doimiysi $k = 1,381 \cdot 10^{-23}$ J/K.

COUNTRY NUMBER	COUNTRY CODE	STUDENT NUMBER	PAGE NUMBER	TOTAL No OF PAGES

Answer sheet 1

TASK 1 (2.0 points)

Draw the electric connections in the boxes and between boxes below.



Photoresistor	
Incandescent Bulb	
Potentiometer	
Red socket	
Black socket	

Ω	Ohmmeter
V	Voltmeter
A	Ammeter
P	Platform
Pm	Potentiometer
B	Battery



COUNTRY NUMBER	COUNTRY CODE	STUDENT NUMBER	PAGE NUMBER	TOTAL No OF PAGES

Answer sheet 2

TASK 2

a) (1.0 points)

$T_0 =$

b) (2.0 points)

V	I	R_B

$$V_{min} = *$$

* This is a characteristic of your apparatus. You can't go below it.

COUNTRY NUMBER	COUNTRY CODE	STUDENT NUMBER	PAGE NUMBER	TOTAL No OF PAGES

Answer sheet 4

TASK 4

a) (2.0 points)

V	I	R

b) (1.5 points)

$R =$	$\gamma =$
$R^* =$	$\Delta\gamma =$

c) (1.0 points)

$$\frac{1}{\Gamma} \frac{d\Gamma}{d\ln\tau} = \frac{1}{\Gamma} \frac{d\Gamma}{d\ln\tau} \quad \text{Eq. (9)}$$

1-MASALA. BAXTSIZ SUN'IY YO'LDOSH YECHIM

1.1 va 1.2

$$\begin{aligned}
 G \frac{M_T m}{r_0^2} &= m \frac{v_0^2}{r_0} \Rightarrow \\
 v_0 &= \frac{2\pi r_0}{T_0} \Rightarrow \\
 g &= \frac{GM_T}{R_T^2} \\
 r_0 &= \left(\frac{g R_T^2 T_0^2}{4\pi^2} \right)^{1/3} \Rightarrow \\
 v_0 &= R_T \sqrt{\frac{g}{v_0}} \Rightarrow \\
 v_0 &= 3,07 \cdot 10^3 \text{ m/s}
 \end{aligned}$$

1.3

$$\begin{aligned}
 L_0 &= r_0 m v_0 = \frac{g R_T^2}{v_0^2} m v_0 \Rightarrow \\
 L_0 &= \frac{m g R_T^2}{v_0} \\
 E_0 &= \frac{1}{2} m v_0^2 - G \frac{M_T m}{r_0} = \frac{1}{2} m v_0^2 - \frac{g R_T^2 m}{r_0} = \frac{1}{2} m v_0^2 - m v_0^2 \Rightarrow \\
 E_0 &= -\frac{1}{2} m v_0^2
 \end{aligned}$$

2.1

Yarim latus rektum l ning qiymati ikkala orbita uchun orbital burchak impulsining bir xilligini hisobga olgan holda topiladi:

$$\begin{aligned}
 l &= \frac{L_0^2}{G M_T m^2} = \frac{m^2 g^2 R_T^4}{v_0^2} \cdot \frac{1}{g R_T^2 m^2} = \frac{g R_T^2}{v_0^2} = r_0 \Rightarrow \\
 l &= r_0
 \end{aligned}$$

Ekssentrisitet qiymati:

$$e^2 = 1 + \frac{2EL_0^2}{G^2 M_T^2 m^3}$$

bu yerda E — sun'iy yo'ldoshning yangi mexanik energiyasi:

$$E = \frac{1}{2} m (v_0^2 + \Delta v^2) - G \frac{M_T m}{r_0} = \frac{1}{2} m \Delta v^2 + E_0 = \frac{1}{2} m \Delta v^2 - \frac{1}{2} m v_0^2$$

ya'ni

$$E = \frac{1}{2}mv_0^2 \left(\frac{\Delta v^2}{v_0^2} - 1 \right) = \frac{1}{2}mv_0^2 (\beta^2 - 1)$$

Ikkalasini birlashtirib, quyidagi kelib chiqadi:

$$e = \beta$$

Bu elliptik traektoriya, chunki $e = \beta < 1$.

2.2

Boshlang'ich va yakuniy orbitalar P nuqtada kesishadi, bu nuqtada sun'iy yo'ldosh dvigateli lahzali ravishda yoqilgan (4-rasmga qarang). Ushbu nuqtada:

$$r(\theta = \alpha) = r_0 = \frac{r_0}{1 - \beta \cos \alpha} \implies \boxed{\alpha = \frac{\pi}{2}}$$

2.3

Traektoriya (trajectory) ifodasidan darhol ko'rinadiki, r ning maksimal va minimal qiymatlari mos ravishda $\theta = 0$ va $\theta = \pi$ burchaklarga to'g'ri keladi (4-rasmga qarang). Shunday qilib, ular quyidagicha beriladi:

$$r_{\max} = \frac{l}{1 - \varepsilon} \quad r_{\min} = \frac{l}{1 + \varepsilon}$$

ya'ni

$$\boxed{r_{\max} = \frac{r_0}{1 - \beta}} \quad \text{va} \quad \boxed{r_{\min} = \frac{r_0}{1 + \beta}}$$

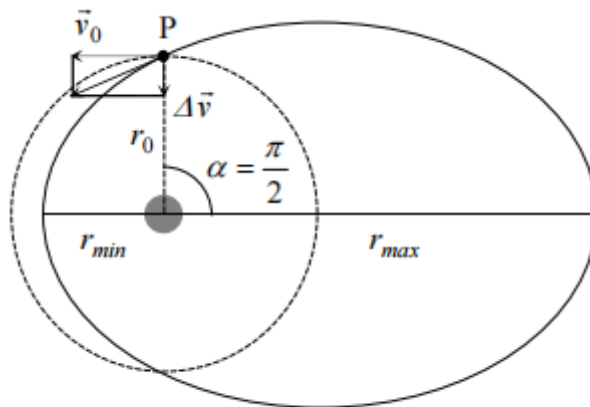


Figure 4

Figure 17: 4-rasm

$\beta = 1/4$ uchun quyidagilar olinadi:

$$r_{\max} = 5,63 \cdot 10^7 \text{ m}; \quad r_{\min} = 3,38 \cdot 10^7 \text{ m}$$

Maksimal masofa $r_{\max}$ va minimal masofa $r_{\min}$ qiymatlarini mexanik energiya va burchakli impuls (angular momentum) saqlanish qonunlaridan ham olish mumkin, bunda apogey va perigey nuqtalarida $\vec{r}$ va $\vec{v}$ vektorlarining o'zaro perpendikulyar ekanligi hisobga olinadi:

$$E = \frac{1}{2}mv_0^2(\beta^2 - 1) = \frac{1}{2}mv^2 - \frac{gR_T^2m}{r}$$

$$L_0 = \frac{mgR_T^2}{v_0} = mvr$$

Ulardan v tezlik chiqarib tashlangandan so'ng, qolgan ifoda yechimlari $r_{\max}$ va $r_{\min}$ bo'lgan ikkinchi darajali tenglama (equation) hisoblanadi.

2.4

Keplerning uchinchi qonuniga ko'ra, yangi orbitadagi davr (period) T quyidagilarni qanoatlantiradi:

$$\frac{T^2}{a^3} = \frac{T_0^2}{r_0^3}$$

bu yerda ellipsning katta yarim o'qi (semi-major axis) a quyidagiga teng:

$$a = \frac{r_{\max} + r_{\min}}{2} = \frac{r_0}{1 - \beta^2}$$

Shuning uchun:

$$T = T_0 (1 - \beta^2)^{-3/2}$$

$\beta = 1/4$ uchun:

$$T = T_0 \left(\frac{15}{16} \right)^{-3/2} = 26,4 \text{ h}$$

3.1

Faqatgina sun'iy yo'ldosh ochiq traektoriya bo'ylab harakatlansa, u Yerning gravitatsion kuch (tortishish kuchi) ta'siridan qutulishi mumkin. U holda, orbita ekscentrisiteti birga teng yoki undan katta bo'lishi kerak. Minimal tezlik oshirilishi parabolik traektoriyaga mos keladi, bunda $\varepsilon = 1$:

$$\varepsilon = \beta \implies \boxed{\beta_{\text{esc}} = 1}$$

Buni, shuningdek, cheksizlikka erishish uchun sun'iy yo'ldoshning to'liq energiyasi nolga teng bo'lishi kerakligidan ($E_p = 0$) hamda qoldiq tezlik bo'lmasligidan ($E_k = 0$) olish mumkin:

$$E = \frac{1}{2}mv_0^2 (\beta_{\text{esc}}^2 - 1) = 0 \implies \beta_{\text{esc}} = 1$$

Bu shuningdan kelib chiqadiki, $T = \infty$ yoki $r_{\text{max}} = \infty$.

3.2

$\varepsilon = \beta_{\text{esc}} = 1$ bo'lganligi sababli, qutb koordinatalaridagi parabola tenglamasi quyidagicha bo'ladi:

$$r = \frac{l}{1 - \cos \theta}$$

bu yerda fokus parametri $l = r_0$ bo'lib qoladi. Minimal Yer-sun'iy yo'ldosh masofasi $\theta = \pi$ burchakka mos keladi, bu yerda:

$$\boxed{r'_{\text{min}} = \frac{r_0}{2}}$$

Bu natija, shuningdek, energiya saqlanish qonunidan ($E = 0$ uchun) va boshlang'ich P nuqtadagi hamda eng yaqin yaqinlashish nuqtasidagi (bu yerda $\vec{r}$ va $\vec{v}$ perpendikulyar) burchakli impuls (L_0) tengligidan kelib chiqadi.

4.1

Agar sun'iy yo'ldosh cheksizlikka v_∞ qoldiq tezlik bilan chiqib ketsa, energiya saqlanish qonuniga ko'ra:

$$E = \frac{1}{2}mv_0^2 (\beta^2 - 1) = \frac{1}{2}mv_\infty^2 \implies$$

$$\boxed{v_\infty = v_0 (\beta^2 - 1)^{1/2}}$$

4.2

$\varepsilon = \beta > \beta_{\text{esc}} = 1$ bo'lganligi sababli, sun'iy yo'ldosh traektoriyasi giperbola bo'ladi.

Sun'iy yo'ldoshning burchakli impulsi P nuqtada, uning qoldiq tezligi v_∞ bo'lgan nuqtadagiga (5-rasmga qarang) qaraganda bir xil bo'ladi, shuning uchun:

$$mv_0 r_0 = mv_\infty b$$

Shunday qilib:

$$b = r_0 \frac{v_0}{v_\infty} \implies \boxed{b = r_0 (\beta^2 - 1)^{-1/2}}$$

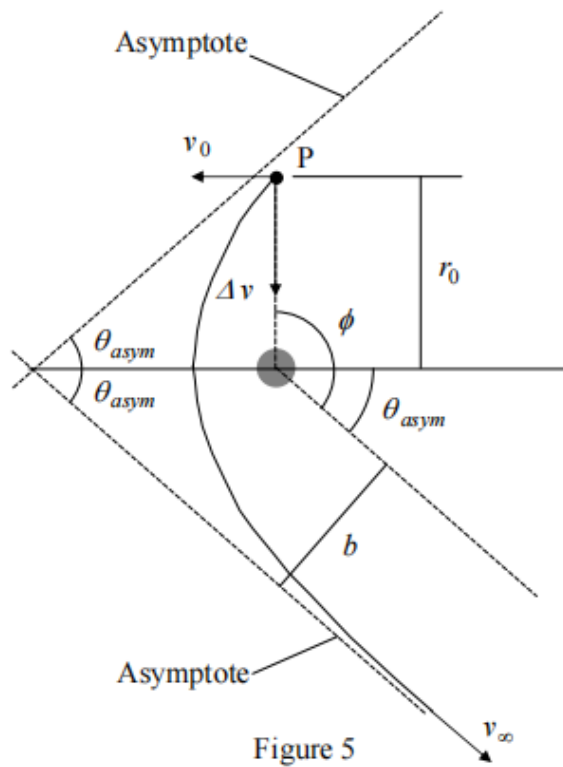


Figure 18: 5-rasm

4.3

Har bir asimptota va giperbola o'qi orasidagi burchak $r \rightarrow \infty$ limitida uning qutb tenglamasida paydo bo'ladigan burchakdir. Bu tenglama maxraji nolga teng bo'ladigan burchakdir:

$$1 - \beta \cos \theta_{\text{asym}} = 0 \implies \theta_{\text{asym}} = \cos^{-1} \left(\frac{1}{\beta} \right)$$

5-rasmga ko'ra:

$$\phi = \frac{\pi}{2} + \theta_{\text{asym}} \implies \boxed{\phi = \frac{\pi}{2} + \cos^{-1} \left(\frac{1}{\beta} \right)}$$

$$\beta = \frac{3}{2} \text{ va } \beta_{\text{esc}} = \frac{3}{2} \text{ uchun quyidagilar olinadi: } \boxed{\phi = 138^\circ = 2,41 \text{ rad}}$$

Th 1 JAVOB VARAQASI (ANSWER SHEET)

Savol	Asosiy formulalar va ishlatilgan g'oyalar	Analitik natijalar	Sonli qiymatlar	Ball ko'rsatkichi
1.1	$G \frac{M_T m}{r_0^2} = m \frac{v_0^2}{r_0}$		$r_0 = 4,22 \cdot 10^7 \text{ m}$	0.3
1.2	$v_0 = \frac{2\pi r_0}{T_0}$ $g = \frac{GM_T}{R_T^2}$	$v_0 = R_T \sqrt{\frac{g}{r_0}}$	$v_0 = 3,07 \cdot 10^3 \text{ m/s}$	0.3 + 0.1
1.3	$\vec{L} = m\vec{r} \times \vec{v}$ $E = \frac{1}{2}mv^2 - G \frac{Mm}{r}$	$L_0 = \frac{mgR_T^2}{v_0}$ $E_0 = -\frac{1}{2}mv_0^2$		0.4 0.4
2.1	Konik kesimlar bo'yicha yordam	$l = r_0$ $\varepsilon = \beta$		0.4 0.5
2.2			$\alpha = \frac{\pi}{2}$	1.0
2.3	2.1 natijalari, yoki E va L ning saqlanishi	$r_{\max} = \frac{r_0}{1-\beta}$ $r_{\min} = \frac{r_0}{1+\beta}$	$r_{\max} = 5,63 \cdot 10^7 \text{ m}$ $r_{\min} = 3,38 \cdot 10^7 \text{ m}$	1.0 + 0.2
2.4	Keplerning uchinchi qonuni	$T = T_0 (1 - \beta^2)^{-3/2}$	$T = 26,4 \text{ h}$	0.5 + 0.2
3.1	$\varepsilon = 1$, $E = 0$, $T = \infty$ yoki $r_{\max} = \infty$		$\beta_{\text{esc}} = 1$	0.5
3.2	$\varepsilon = 1$ va 2.1 natijalari	$r'_{\min} = \frac{r_0}{2}$		1.0
4.1	Energiya saqlanish qonuni	$v_{\infty} = v_0 (\beta^2 - 1)^{1/2}$		1.0
4.2	Burchakli impuls saqlanishi	$b = r_0 (\beta^2 - 1)^{-1/2}$		1.0
4.3	Konik kesimlar bo'yicha yordam	$\phi = \frac{\pi}{2} + \cos^{-1} \left(\frac{1}{\beta} \right)$	$\phi = 138^\circ = 2,41 \text{ rad}$	1.0 + 0.2

Th 1 Solution Page 5 of 5

36th International Physics Olympiad. Salamanca (España) 2005

Th 2 ABSOLUTE MEASUREMENTS OF ELECTRICAL QUANTITIES SOLUTION

1.

Ma'lum bir t vaqtdan keyin, g'altak tekisligiga o'tkazilgan normal $\vec{B}_0 = B_0 \vec{i}$ magnit maydoni bilan ωt burchak hosil qiladi. U holda g'altak orqali o'tayotgan magnit oqimi:

$$\phi = N \vec{B}_0 \cdot \vec{S}$$

bu yerda $\vec{S}$ sirt vektori quyidagicha beriladi: $\vec{S} = \pi a^2 (\cos \omega t \vec{i} + \sin \omega t \vec{j})$.

Shuning uchun:

$$\phi = N \pi a^2 B_0 \cos \omega t$$

Induksiyalangan elektr yurituvchi kuch (e.yu.k.):

$$\mathcal{E} = -\frac{d\phi}{dt} \implies \boxed{\mathcal{E} = N \pi a^2 B_0 \omega \sin \omega t}$$

Oniy quvvat $P = \mathcal{E}^2/R$ ga teng, shuning uchun:

$$\langle P \rangle = \frac{(N\pi a^2 B_0 \omega)^2}{2R}$$

bu yerda $\langle \sin^2 \omega t \rangle = \frac{1}{T} \int_0^T \sin^2 \omega t dt = \frac{1}{2}$ munosabatdan foydalanildi.

2.

G'altak markazidagi to'liq maydon t vaqt momentida:

$$\vec{B}_t = \vec{B}_0 + \vec{B}_i$$

bu yerda $\vec{B}_i$ induksiyalangan tok hosil qilgan magnit maydoni: $\vec{B}_i = B_i (\cos \omega t \vec{i} + \sin \omega t \vec{j})$,

bunda:

$$B_i = \frac{\mu_0 N I}{2a} \quad \text{va} \quad I = \mathcal{E}/R$$

Shuning uchun:

$$B_i = \frac{\mu_0 N^2 \pi a B_0 \omega}{2R} \sin \omega t$$

Uning komponentlarining o'rtacha qiymatlari:

$$\begin{aligned} \langle B_{ix} \rangle &= \frac{\mu_0 N^2 \pi a B_0 \omega}{2R} \langle \sin \omega t \cos \omega t \rangle = 0 \\ \langle B_{iy} \rangle &= \frac{\mu_0 N^2 \pi a B_0 \omega}{2R} \langle \sin^2 \omega t \rangle = \frac{\mu_0 N^2 \pi a B_0 \omega}{4R} \end{aligned}$$

To'liq magnit maydonining o'rtacha qiymati esa:

$$\langle \vec{B}_t \rangle = B_0 \vec{i} + \frac{\mu_0 N^2 \pi a B_0 \omega}{4R} \vec{j}$$

Magnit mili (strelka) o'rtacha maydon yo'nalishi bo'ylab orientatsiyalanadi, shuning uchun:

$$\tan \theta = \frac{\mu_0 N^2 \pi a \omega}{4R}$$

Natijada, ushbu usul orqali o'lchangan g'altakning qarshiligi θ orqali quyidagicha ifodalanadi:

$$R = \frac{\mu_0 N^2 \pi a \omega}{4 \tan \theta}$$

3.

Diskdagi birlik musbat zaryadga ta'sir qiluvchi kuch radial yo'nalgan va uning moduli:

$$|\vec{v} \times \vec{B}| = vB = \omega r B$$

bu yerda B — g'altak markazidagi magnit maydoni:

$$B = N \frac{\mu_0 I}{2a}$$

U holda, magnit maydon B tomonidan har bir diskda induksiyalangan elektr yurituvchi kuch (e.yu.k.):

$$\mathcal{E}_D = \mathcal{E}_{D'} = B\omega \int_0^b r dr = \frac{1}{2} B\omega b^2$$

Nihoyat, 1 va 4 nuqtalar orasidagi to'liq induksiyalangan e.yu.k. $\mathcal{E} = \mathcal{E}_D + \mathcal{E}_{D'}$ ga teng:

$$\boxed{\mathcal{E} = N \frac{\mu_0 b^2 \omega I}{2a}}$$

4.

Galvanometr G ning ko'rsatkichi nolga teng bo'lganda, $I_G = 0$ bo'ladi va Kirxgof qonunlari darhol javobni beradi. U holda bizda:

$$\mathcal{E} = IR \implies \boxed{R = N \frac{\mu_0 b^2 \omega}{2a}}$$

5.

h masofa bilan ajratilgan ikkita cheksiz to'g'ri parallel o'tkazgichlar orasidagi birlik uzunlikka to'g'ri keladigan o'zaro ta'sir kuchi f :

$$f = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{h}$$

$I_1 = I_2 = I$ bo'lganda va $2\pi a$ uzunlik uchun, qo'shni C_1 g'altaklar tomonidan C_2 g'altakka ta'sir etuvchi induksiyalangan kuch F :

$$\boxed{F = \frac{\mu_0 a}{h} I^2}$$

6.

Muvozanat holatida:

$$mgx = 4Fd$$

U holda:

$$mgx = \frac{4\mu_0 ad}{h} I^2 \quad (1)$$

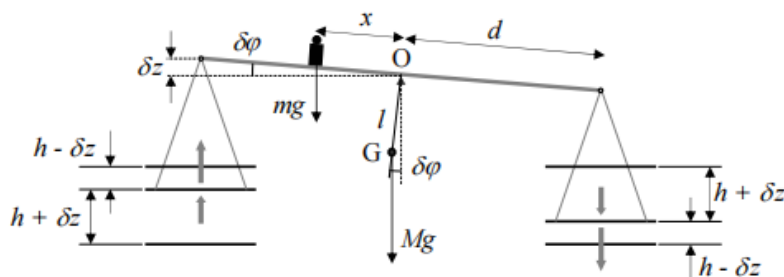
bu yerdan:

$$\boxed{I = \left(\frac{mgx}{4\mu_0 ad} \right)^{1/2}}$$

7.

Agar tayanch nuqtasi O ga nisbatan og'irlik kuchi momentlari magnit kuchlar momentlaridan katta bo'lsa, tarozi kichik $\delta\varphi$ burchakli og'ishdan so'ng yana muvozanat holatiga qaytadi:

$$Mgl \sin \delta\varphi + mgx \cos \delta\varphi > 2\mu_0 a I^2 \left(\frac{1}{h - \delta z} + \frac{1}{h + \delta z} \right) d \cos \delta\varphi$$



Shuning uchun, taklif qilingan yaqinlashishdan foydalanib:

$$Mgl \sin \delta\varphi + mgx \cos \delta\varphi > \frac{4\mu_0 a d I^2}{h} \left(1 + \frac{\delta z^2}{h^2} \right) \cos \delta\varphi$$

(1) muvozanat shartini hisobga olgan holda quyidagiga ega bo'lamiz:

$$Mgl \sin \delta\varphi > mgx \frac{\delta z^2}{h^2} \cos \delta\varphi$$

Nihoyat, $\tan \delta\varphi \approx \sin \delta\varphi = \frac{\delta z}{d}$ bo'lganligi uchun:

$$\delta z < \frac{Mlh^2}{mxd} \implies \boxed{\delta z_{\max} = \frac{Mlh^2}{mxd}}$$

Solution Th 2 Page 3 of 4

Th 2 JAVOB VARAQASI (ANSWER SHEET)

Savol	Asosiy formulalar va ishlatilgan g'oyalar	Analitik natijalar	Ball
1	$\Phi = N\vec{B}_0 \cdot \vec{S}$	$\mathcal{E} = N\pi a^2 B_0 \omega \sin \omega t$	0.5
	$\mathcal{E} = -\frac{d\Phi}{dt}$ $P = \frac{\mathcal{E}^2}{R}$	$\langle P \rangle = \frac{(N\pi a^2 B_0 \omega)^2}{2R}$	1.0
2	$\vec{B} = \vec{B}_0 + \vec{B}_i$	$R = \frac{\mu_0 N^2 \pi a \omega}{4 \tan \theta}$	2.0
	$B_i = \frac{\mu_0 N}{2a} I$ $\tan \theta = \frac{\langle B_y \rangle}{\langle B_x \rangle}$		
3	$\vec{E} = \vec{v} \times \vec{B}$	$\mathcal{E} = N \frac{\mu_0 b^2 \omega I}{2a}$	2.0
	$v = \omega r$ $B = N \frac{\mu_0 I}{2a}$ $\mathcal{E} = \int_0^b \vec{E} d\vec{r}$		
4	$\mathcal{E} = RI$	$R = N \frac{\mu_0 b^2 \omega}{2a}$	0.5
5	$f = \frac{\mu_0}{2\pi} \frac{II'}{h}$	$F = \frac{\mu_0 a}{h} I^2$	1.0
6	$mgx = 4Fd$	$I = \left(\frac{mghx}{4\mu_0 ad} \right)^{1/2}$	1.0
7	$\Gamma_{\text{grav}} > \Gamma_{\text{mag}}$	$\delta z_{\text{max}} = \frac{Mlh^2}{mxd}$	2.0

Solution Th 2 Page 4 of 4

Th 3 QUANTUM EFFECTS OF GRAVITY SOLUTION

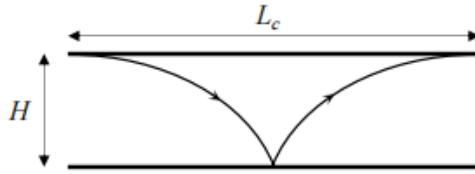
1.

A yutilish sirtida faqatgina H balandlikdan o'ta olmaydigan neytronlarga omon qoladi. Ularning qayrilish (burilish) nuqtalari H dan pastda bo'ladi. Shunday qilib, bo'shliqqa z balandlikda v_z vertikal tezlik bilan kiradigan neytron uchun energiya saqlanish qonuni quyidagicha yoziladi:

$$\frac{1}{2}Mv_z^2 + Mgz \leq Mgh \implies \boxed{-\sqrt{2g(H-z)} \leq v_z(z) \leq \sqrt{2g(H-z)}}$$

2.

Ruxsat etilgan diapazondan tashqaridagi tezlikka ega bo'lgan barcha neytronlarning to'liq yutilishini ta'minlash uchun kamera yetarlicha uzun bo'lishi kerak. Shuning uchun, neytronlar kamera ichida kamida bir marta o'zlarining maksimal balandliklariga erishishlari kerak. Eng uzun talab qilinadigan uzunlik $z = H$ balandlikda $v_z = 0$ bilan kiradigan neytronlarga mos keladi (rasmga qarang). Ularning tushish vaqtini t_f deb belgilasak:



$$\left. \begin{aligned} L_c &= v_x 2t_f \\ H &= \frac{1}{2}gt_f^2 \end{aligned} \right\} \implies \boxed{L_c = 2v_x \sqrt{\frac{2H}{g}}} \quad \boxed{L_c = 6.4 \text{ cm}}$$

3.

Berilgan z balandlikda kiruvchi neytronlarning birlik balandlikka to'g'ri keladigan o'tish jadalligi (rate of transmitted neutrons) shu balandlikdagi ruxsat etilgan tezliklar oralig'iga proporsionaldir, bunda proporsionallik koeffitsiyenti ρ :

$$\frac{dN_c(z)}{dz} = \rho [v_{z,\max}(z) - v_{z,\min}(z)] = 2\rho\sqrt{2g(H-z)}$$

O'tgan neytronlarning umumiy soni barcha mumkin bo'lgan balandliklarda kiradigan neytronlarni qo'shish (integrallash) orqali olinadi. $y = z/H$ deb belgilab:

$$\begin{aligned} N_c(H) &= \int_0^H dN_c(z) = \int_0^H 2\rho\sqrt{2g(H-z)} dz = 2\rho\sqrt{2g} H^{3/2} \int_0^1 (1-y)^{1/2} dy \\ &= 2\rho\sqrt{2g} H^{3/2} \left[-\frac{2}{3}(1-y)^{3/2} \right]_0^1 \implies \boxed{N_c(H) = \frac{4}{3}\rho\sqrt{2g} H^{3/2}} \end{aligned}$$

4.

H balandlikdan tushayotgan neytron uchun, sakrash siklidagi ta'sir (action) tushish yoki ko'tarilish vaqtidagi ta'sirdan ikki baravar ko'pdir:

$$S = 2 \int_0^H p_z dz = 2M\sqrt{2g} H^{3/2} \int_0^1 (1-y)^{1/2} dy = \frac{4}{3} M\sqrt{2g} H^{3/2}$$

Bor-Zommerfeld (BS) kvantlash shartidan foydalanib:

$$S = \frac{4}{3} M\sqrt{2g} H^{3/2} = nh \implies H_n = \left(\frac{9h^2}{32M^2g} \right)^{1/3} n^{2/3}$$

Unga mos keladigan energiya sathlari (vertikal harakat bilan bog'liq):

$$E_n = MgH_n \implies E_n = \left(\frac{9Mg^2h^2}{32} \right)^{1/3} n^{2/3}$$

Th 3 Solution Page 1 of 3

Birinchi sath uchun sonli qiymatlar:

$$H_1 = \left(\frac{9h^2}{32M^2g} \right)^{1/3} = 1,65 \cdot 10^{-5} \text{ m} \quad \boxed{H_1 = 16,5 \mu\text{m}}$$

$$E_1 = MgH_1 = 2,71 \cdot 10^{-31} \text{ J} = 1,69 \cdot 10^{-12} \text{ eV} \quad \boxed{E_1 = 1,69 \text{ peV}}$$

E'tibor bering, H_1 olingan kamera balandligi $H = 50 \mu\text{m}$ bilan bir xil tartibda (tartibdosh) ekanligini ko'rish mumkin. Bu esa H ni o'zgartirish orqali fazoviy kvantlanishni kuzatish imkoniyatini ochib beradi.

5.

Noaniqlik prinsipiga ko'ra, minimal vaqt Δt va minimal energiya ΔE quyidagi munosabatni qanoatlantiradi: $\Delta E \Delta t \geq \hbar$. Ushbu vaqt davomida neytronlar o'ngga quyidagi masofaga siljiydi:

$$\Delta x = v_x \Delta t \geq v_x \frac{\hbar}{\Delta E}$$

Kameraga ruxsat etilgan neytronning minimal energiyasi E_1 ga teng, shuning uchun $\Delta E \approx E_1$. Demak, talab qilinadigan minimal vaqt va minimal uzunlikning bahosi quyidagicha bo'ladi:

$$t_q \approx \frac{\hbar}{E_1} = 0,4 \cdot 10^{-3} \text{ s} = 0,4 \text{ ms} \quad L_q \approx v_x \frac{\hbar}{E_1} = 4 \cdot 10^{-3} \text{ m} = 4 \text{ mm}$$

Th 3 Solution Page 2 of 3

Th 3 JAVOB VARAQASI (ANSWER SHEET)

Savol	Asosiy formulalar va g'oyalar	Analitik natijalar	Sonli natijalar	Ball
1	$\frac{1}{2}Mv_z^2 + Mgz \leq Mgh$	$-\sqrt{2g(H-z)} \leq v_z(z) \leq \sqrt{2g(H-z)}$		1.5
2	$L_c = v_x 2t_f$ $H = \frac{1}{2}gt_f^2$	$L_c = 2v_x \sqrt{\frac{2H}{g}}$	$L_c = 6.4 \text{ cm}$	1.3 + 0.2
3	$\frac{dN_c}{dz} = \rho [v_{z,\max} - v_{z,\min}]$ $N_c(H) = \int_0^H dN_c(z)$	$N_c(H) = \frac{4}{3}\rho\sqrt{2g} H^{3/2}$		2.5
4	$S = 2 \int_0^H p_z dz = nh$	$H_n = \left(\frac{9h^2}{32M^2g}\right)^{1/3} n^{2/3}$ $E_n = \left(\frac{9Mg^2h^2}{32}\right)^{1/3} n^{2/3}$	$H_1 = 16,5 \mu\text{m}$ $E_1 = 1,69 \text{ peV}$	1.6 + 0.2 0.5 + 0.2
5	$\Delta E \Delta t \geq \hbar$ $\Delta E \approx E_1$ $\Delta x = v_x \Delta t$	$t_q \approx \frac{\hbar}{E_1}$ $L_q \approx v_x \frac{\hbar}{E_1}$	$t_q \approx 0,4 \text{ ms}$ $L_q \approx 4 \text{ mm}$	1.3 + 0.2 0.3 + 0.2

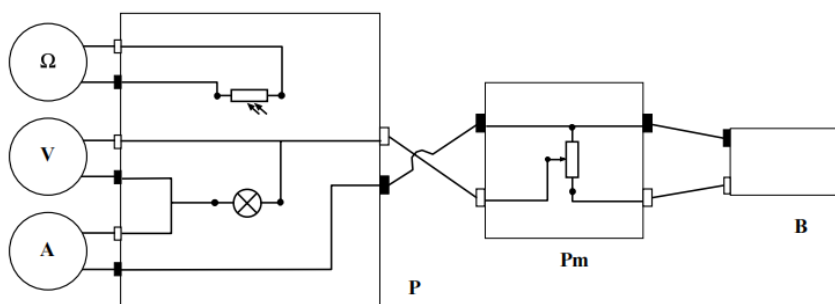
Th 3 Solution Page 3 of 3

PLANCK'S CONSTANT IN THE LIGHT OF AN INCANDESCENT LAMP SOLUTION

TASK 1

Quyidagi qutilarda va qutilar orasidagi elektr ulanishlarni chizing (Sxema rasmda ko'rsatilganidek amalga oshiriladi).

Photoresistor	Fotorezistor	Ω	Ommetr
Incandescent Bulb	Cho'g'lanma lamba	V	Voltmetr
Potentiometer	Potensiometr	A	Ampermetr
Red socket	Qizil rozetka	P	Platforma
Black socket	Qora rozetka	Pm	Potensiometr
		B	Batareya



Photoresistor		Ω	Ohmmeter
Incandescent Bulb		V	Voltmeter
Potentiometer		A	Ammeter
Red socket		P	Platform
Black socket		Pm	Potentiometer
		B	Battery

TASK 2

a)

$$t_0 = 24^\circ\text{C} \quad T_0 = 297\text{ K} \quad \Delta T_0 = 1\text{ K}$$

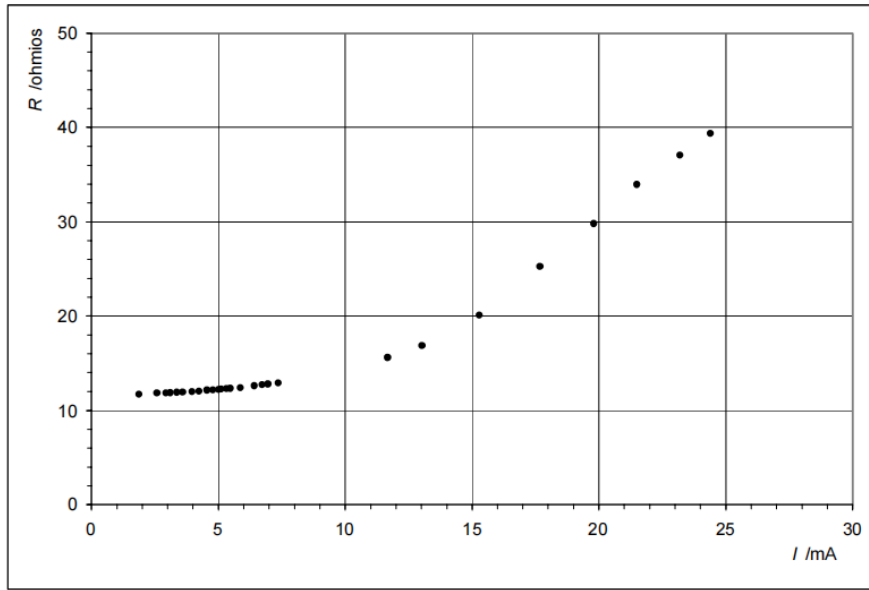
b)

V/mV	I/mA	R_B/Ω	V/mV	I/mA	R_B/Ω
21.9	1.87	11.7	61.3	5.02	12.2
30.5	2.58	11.8	65.5	5.33	12.3
34.9	2.95	11.8	67.5	5.47	12.3
37.0	3.12	11.9	73.0	5.88	12.4
40.1	3.37	11.9	80.9	6.42	12.6
43.0	3.60	11.9	85.6	6.73	12.7
47.6	3.97	12.0	89.0	6.96	12.8
51.1	4.24	12.1	95.1	7.36	12.9
55.3	4.56	12.1	111.9	8.38	13.4
58.3	4.79	12.2	130.2	9.37	13.9

$$V_{\min} = 9.2 \text{ mV} \quad *$$

* *Bu qurilmangizning o'ziga xos xarakteristikasidir. Undan pastga tusha olmaysiz.*

Biz vertikal o'qda R_B ning qiymatlarini I ga bog'liqlik grafigini chizamiz. R_{B0} ni hisoblash uchun biz birinchi o'nta ko'rsatkichni tanlaymiz.



c)

V/mV	I/mA	R _B /Ω
21.9 ± 0.1	1.87 ± 0.01	11.7 ± 0.1
30.5 ± 0.1	2.58 ± 0.01	11.8 ± 0.1
34.9 ± 0.1	2.95 ± 0.01	11.8 ± 0.1
37.0 ± 0.1	3.12 ± 0.01	11.9 ± 0.1
40.1 ± 0.1	3.37 ± 0.01	11.9 ± 0.1
43.0 ± 0.1	3.60 ± 0.01	11.9 ± 0.1
47.6 ± 0.1	3.97 ± 0.01	12.0 ± 0.1
51.1 ± 0.1	4.24 ± 0.01	12.1 ± 0.1
55.3 ± 0.1	4.56 ± 0.01	12.1 ± 0.1
58.3 ± 0.1	4.79 ± 0.01	12.2 ± 0.1

Exp. Page 3 of 9

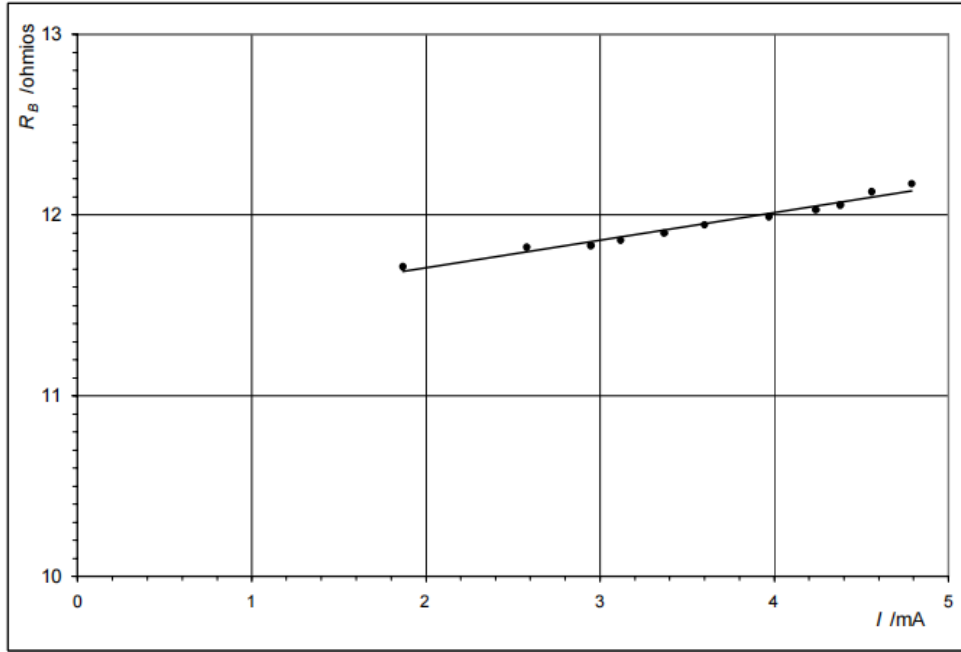


Figure 19: 1-rasm: R_B ning I tokka bog'liqlik grafigi

R_B uchun xatolik (Misol tariqasida, birinchi qiymat uchun xatolikni hisoblaymiz).

$$\Delta R_B = R_B \sqrt{\left(\frac{\Delta V}{V}\right)^2 + \left(\frac{\Delta I}{I}\right)^2} = 11,71 \sqrt{\left(\frac{0,1}{21,9}\right)^2 + \left(\frac{0,01}{1,87}\right)^2} = 0,1$$

Biz eng kichik kvadratlar usuli orqali R_{B0} qiymatini hisoblab chiqdik.

$$R_{B0} = 11,4$$

$$\text{qiya tekislik burchagi} = m = 0,167$$

$$\sum I^2 = 130,38$$

$$\sum I = 35,05$$

$$n = 10$$

$$X \text{ o'qi uchun: } \sigma_I = \sqrt{\frac{\sum \Delta I^2}{n}} = 0,01$$

$$Y \text{ o'qi uchun: } \sigma_{R_B} = \sqrt{\frac{\sum \Delta R_B^2}{n}} = 0,047$$

$$\sigma = \sqrt{\sigma_{R_B}^2 + m^2 \sigma_I^2} = \sqrt{0,01^2 + 0,167^2 \cdot 0,01^2} = 0,1$$

$$\Delta R_{B0} = \sqrt{\frac{\sigma^2 \sum I^2}{n \sum I^2 - (\sum I)^2}} = \sqrt{\frac{0,1^2 \cdot 130,38}{10 \cdot 130,38 - 35,05^2}} = 0,13$$

$R_{B0} = 11,4 \Omega$	$\Delta R_{B0} = 0,1 \Omega$
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d) $T = aR^{0.83}$; $a = \frac{T_0}{R_0^{0.83}}$; $a = \frac{297}{11.4^{0.83}} = 39.40$

Ikkiti usul uchun xatolikni hisoblash (Working out the error for two methods):

A usuli (Method A)

$$\ln a = \ln T_0 - 0.83 \ln R_{B0} \quad ; \quad \Delta a = a \left(\frac{\Delta T_0}{T_0} + 0.83 \frac{\Delta R_{B0}}{R_{B0}} \right) \quad ; \quad \Delta a = 39.40 \left(\frac{1}{297} + 0.83 \frac{0.1}{11.40} \right) = 0.419 = 0.4$$

B usuli (Method B)

a ning kattaroq qiymati (Higher value of a):

$$a_{\max} = \frac{T_0 + \Delta T_0}{(R_0 - \Delta R_0)^{0.83}} = \frac{297 + 1}{(11.4 - 0.1)^{0.83}} = 39.8255$$

a ning kichikroq qiymati (Smaller value of a):

$$a_{\min} = \frac{T_0 - \Delta T_0}{(R_0 + \Delta R_0)^{0.83}} = \frac{297 - 1}{(11.4 + 0.1)^{0.83}} = 38.9863$$

$$\Delta a = \frac{a_{\max} - a_{\min}}{2} = \frac{39.8255 - 38.9863}{2} = 0.419 = 0.4$$

$a = 39.4$	$\Delta a = 0.4$
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3-VAZIFA (TASK 3)

Chunki (Because of) $2\Delta\lambda = 620 - 565$; $\Delta\lambda = 28 \text{ nm}$

$\lambda_0 = 590 \text{ nm}$	$\Delta\lambda = 28 \text{ nm}$
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4-VAZIFA (TASK 4)

a)

V/V	I/mA	R/kΩ
9.48	85.5	8.77
9.73	86.8	8.11
9.83	87.3	7.90
100.1	88.2	7.49
10.25	89.4	7.00
10.41	90.2	6.67
10.61	91.2	6.35
10.72	91.8	6.16
10.82	92.2	6.01
10.97	93.0	5.77
11.03	93.3	5.69
11.27	94.5	5.35
11.42	95.1	5.17
11.50	95.5	5.07

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b)

Chunki (Because of) $\ln \frac{R}{R'} = \gamma \ln 0.512$; $\gamma = \ln \frac{R}{R'} / \ln 0.512 = \ln \frac{5.07}{8.11} / \ln 0.512 = 0.702$

$\Delta\gamma$ ni hisoblash uchun bizga ma'lumki (For working out $\Delta\gamma$ we know that):

$$R \pm \Delta R = 5.07 \pm 0.01 \text{ k}\Omega$$

$$R' \pm \Delta R' = 8.11 \pm 0.01 \text{ k}\Omega$$

O'tkazuvchanlik (Transmittance), $t = 51.2\%$

Ikkita usul uchun xatolikni hisoblash (Working out the error for two methods):

A usuli (Method A)

$$\gamma = \frac{\ln R/R'}{\ln t} ; \quad \Delta\gamma = \frac{1}{\ln t} \left(\frac{\Delta R}{R} + \frac{\Delta R'}{R'} \right) = \frac{1}{\ln 0.512} \left(\frac{0.01}{5.07} + \frac{0.01}{8.11} \right) = 0.00479 ; \quad \Delta\gamma = 0.005$$

B usuli (Method B)

γ ning kattaroq qiymati (Higher value of γ):

$$\gamma_{\max} = \ln \frac{R - \Delta R}{R' + \Delta R'} / \ln t = \ln \frac{5.07 - 0.01}{8.11 + 0.01} / \ln 0.512 = 0.70654$$

γ ning kichikroq qiymati (Smaller value of γ):

$$\gamma_{\min} = \ln \frac{R + \Delta R}{R' - \Delta R'} / \ln t = \ln \frac{5.07 + 0.01}{8.11 - 0.01} / \ln 0.512 = 0.69696$$

$$\Delta\gamma = \frac{\gamma_{\max} - \gamma_{\min}}{2} = \frac{0.70654 - 0.69696}{2} = 0.00479 \quad ; \quad \Delta\gamma = 0.005$$

$R = 5.07 \text{ k}\Omega$	$\gamma = 0.702$
$R' = 8.11 \text{ k}\Omega$	$\Delta\gamma = 0.005$

c)

Bizga ma'lumki (We know that)	$R = c_3 e^{\frac{c_2 \gamma}{\lambda_0 T}} (3)$
u holda (then)	$\ln R = \ln c_3 + \frac{c_2 \gamma}{\lambda_0 T}$
Chunki (Because of)	$T = a R_B^{0.83} (6)$
natijada (consequently)	$\ln R = \ln c_3 + \frac{c_2 \gamma}{\lambda_0 a} R_B^{-0.83}$

$$\ln R = \ln c_3 + \frac{c_2 \gamma}{\lambda_0 a} R_B^{-0.83} \quad \text{Eq. (9)}$$

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d)

V/V	I/mA	R_B/Ω	T/K	$R_B^{-0.83} \text{ (S.I.)}$	$R/\text{k}\Omega$	$\ln R$
9.48 ± 0.01	85.5 ± 0.1	110.9 ± 0.2	1962 ± 18	$(2.008 \pm 0.004) \times 10^{-2}$	8.77 ± 0.01	2.171 ± 0.001
9.73 ± 0.01	86.8 ± 0.1	112.1 ± 0.2	1980 ± 18	$(1.990 \pm 0.004) \times 10^{-2}$	8.11 ± 0.01	2.093 ± 0.001
9.83 ± 0.01	87.3 ± 0.1	112.6 ± 0.2	1987 ± 18	$(1.983 \pm 0.004) \times 10^{-2}$	7.90 ± 0.01	2.067 ± 0.001
10.01 ± 0.01	88.2 ± 0.1	113.5 ± 0.2	2000 ± 18	$(1.970 \pm 0.004) \times 10^{-2}$	7.49 ± 0.01	2.014 ± 0.001
10.25 ± 0.01	89.4 ± 0.1	114.7 ± 0.2	2018 ± 18	$(1.952 \pm 0.003) \times 10^{-2}$	7.00 ± 0.01	1.946 ± 0.001
10.41 ± 0.01	90.2 ± 0.1	115.4 ± 0.2	2028 ± 18	$(1.943 \pm 0.003) \times 10^{-2}$	6.67 ± 0.01	1.894 ± 0.002
10.61 ± 0.01	91.2 ± 0.1	116.3 ± 0.2	2041 ± 18	$(1.930 \pm 0.003) \times 10^{-2}$	6.35 ± 0.01	1.849 ± 0.002
10.72 ± 0.01	91.8 ± 0.1	116.8 ± 0.2	2049 ± 19	$(1.923 \pm 0.003) \times 10^{-2}$	6.16 ± 0.01	1.818 ± 0.002
10.82 ± 0.01	92.2 ± 0.1	117.4 ± 0.2	2057 ± 19	$(1.915 \pm 0.003) \times 10^{-2}$	6.01 ± 0.01	1.793 ± 0.002
10.97 ± 0.01	93.0 ± 0.1	118.0 ± 0.2	2066 ± 19	$(1.907 \pm 0.003) \times 10^{-2}$	5.77 ± 0.01	1.753 ± 0.002
11.03 ± 0.01	93.3 ± 0.1	118.2 ± 0.2	2069 ± 19	$(1.904 \pm 0.003) \times 10^{-2}$	5.69 ± 0.01	1.739 ± 0.002
11.27 ± 0.01	94.5 ± 0.1	119.3 ± 0.2	2085 ± 19	$(1.890 \pm 0.003) \times 10^{-2}$	5.35 ± 0.01	1.677 ± 0.002
11.42 ± 0.01	95.1 ± 0.1	120.1 ± 0.2	2096 ± 19	$(1.880 \pm 0.003) \times 10^{-2}$	5.15 ± 0.01	1.639 ± 0.002
11.50 ± 0.01	95.5 ± 0.1	120.4 ± 0.2	2101 ± 19	$(1.875 \pm 0.003) \times 10^{-2}$	5.07 ± 0.01	1.623 ± 0.002

ortiqcha (unnecessary)

Misol tariqasida, biz birinchi qator uchun xatoliklarni hisoblaymiz (We work out the errors for all the first row, as example).

R_B uchun xatolik (Error for R_B):

$$\Delta R_B = R_B \sqrt{\left(\frac{\Delta V}{V}\right)^2 + \left(\frac{\Delta I}{I}\right)^2} = 110.9 \sqrt{\left(\frac{0.01}{9.48}\right)^2 + \left(\frac{0.1}{85.5}\right)^2} = 0.2 \Omega$$

T uchun xatolik (Error for T):

$$\Delta T = T \left(\frac{\Delta a}{a} + 0.83 \frac{\Delta R_B}{R_B} \right) \quad ; \quad \Delta T = 1962 \left(\frac{0.3}{39.4} + 0.83 \frac{0.2}{110.9} \right) = 18 \text{ K}$$

$R_B^{-0.83}$ uchun xatolik (Error for $R_B^{-0.83}$):

$$x = R_B^{-0.83} \quad ; \quad \ln x = -0.83 \ln R_B \quad ; \quad \Delta x = x \cdot 0.83 \frac{\Delta R_B}{R_B} \quad ; \quad \Delta (R_B^{-0.83}) = R_B^{-0.83} \frac{\Delta R_B}{R_B}$$

$$\Delta (R_B^{-0.83}) = 0.020077 \frac{0.2}{110.9} = 0.004 \times 10^{-2}$$

$\ln R$ uchun xatolik (Error for $\ln R$):

$$\Delta \ln R = \frac{\Delta R}{R} \quad ; \quad \Delta \ln R = \frac{0.01}{8.77} = 0.001$$

e)

Biz $\ln R$ ning $R_B^{-0.83}$ ga bog'liqlik grafigini chizamiz (We plot $\ln R$ versus $R_B^{-0.83}$).

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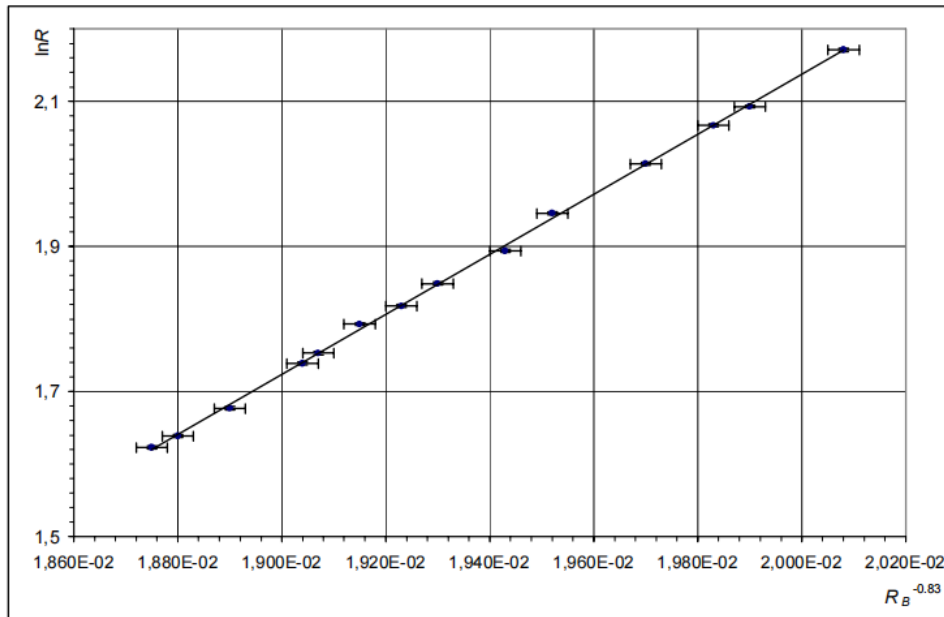


Figure 20: 1-rasm: $\ln R$ ning $R_B^{-0.83}$ ga bog'liqlik grafigi

Eng kichik kvadratlar usuli bo'yicha (By the least squares):
Nishablik (Slope) = $m = 414.6717$

$$\begin{aligned}\sum [R_B^{-0.83}]^2 &= 5.23559 \times 10^{-3} \\ \sum [R_B^{-0.83}] &= 0.27068 \\ n &= 14\end{aligned}$$

$$X \text{ o'qi uchun (For axis } X): \sigma_{R_B^{-0.83}} = \sqrt{\frac{\sum [\Delta R_B^{-0.83}]^2}{n}} = 0.003 \times 10^{-2}$$

$$Y \text{ o'qi uchun (For axis } Y): \sigma_{\ln R} = \sqrt{\frac{\sum [\Delta \ln R]^2}{n}} = 0.002$$

$$\sigma = \sqrt{\sigma_{\ln R}^2 + m^2 \sigma_{R_B^{-0.83}}^2} = \sqrt{0.002^2 + 414.672^2 \cdot [0.003 \times 10^{-2}]^2} = 0.0126$$

$$\Delta m = \sqrt{\frac{n\sigma^2}{n \sum [R_B^{-0.83}]^2 - [\sum R_B^{-0.83}]^2}} = \sqrt{\frac{14 \cdot 0.0126^2}{14 \cdot 5.23559 \times 10^{-3} - [0.27068]^2}} = 8.295$$

Chunki (Because of)

$$m = \frac{c_2 \gamma}{\lambda_0 a}$$

va (and)

$$c_2 = \frac{hc}{k}$$

u holda (then)

$$h = \frac{mk\lambda_0 a}{c\gamma}$$

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$$h = \frac{414.67 \cdot 1.381 \times 10^{-23} \cdot 590 \times 10^{-9} \cdot 39.4}{2.998 \times 10^8 \cdot 0.702} = 6.33 \times 10^{-34}$$

$$\Delta h = h \sqrt{\left(\frac{\Delta m}{m}\right)^2 + \left(\frac{\Delta k}{k}\right)^2 + \left(\frac{\Delta \lambda_0}{\lambda_0}\right)^2 + \left(\frac{\Delta a}{a}\right)^2 + \left(\frac{\Delta \gamma}{\gamma}\right)^2}$$

$$\Delta h = 6.34 \times 10^{-34} \sqrt{\left(\frac{8.3}{415}\right)^2 + 0 + \left(\frac{28}{590}\right)^2 + \left(\frac{0.3}{39.4}\right)^2 + 0 + \left(\frac{0.01}{0.70}\right)^2} = 0.34 \times 10^{-34}$$

$h = 6.3 \times 10^{-34} \text{ J} \cdot \text{s}$	$\Delta h = 0.3 \times 10^{-34} \text{ J} \cdot \text{s}$
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